

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

Noise

Families

Adaptive

The fin

Fields

Representing Functions and Fields

Finitely many numbers for a function, and what each choice loses

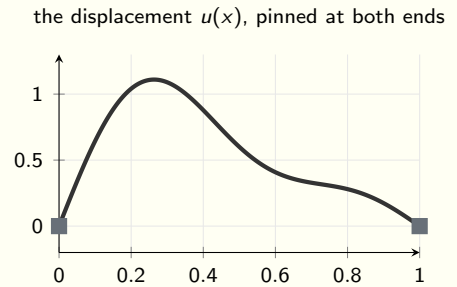
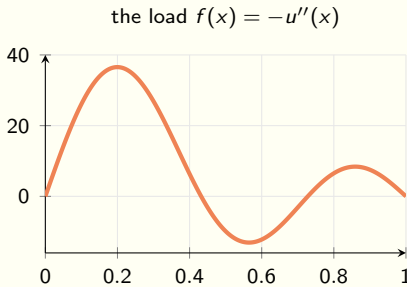
Krishna Kumar

University of Texas at Austin

krishnak@utexas.edu

A taut string under a load, $\mathcal{L}u = f$

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



$$-\frac{d^2 u}{dx^2} = f(x), \quad u(0) = u(1) = 0, \quad \mathcal{L} = -\frac{d^2}{dx^2}$$
$$u(x) = 0.8 \sin(\pi x) + 0.4 \sin(2\pi x) + 0.2 \sin(3\pi x)$$

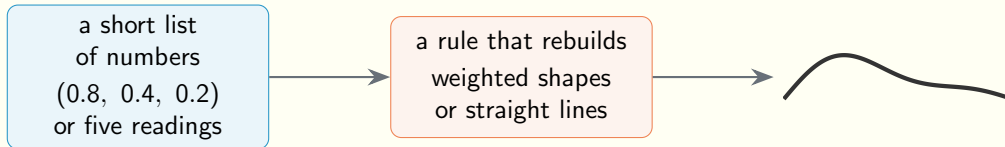
Consider a taut string pinned at $x = 0$ and $x = 1$. A distributed load $f(x)$ deflects the string and produces a displacement $u(x)$, and for small slopes and unit tension the displacement satisfies the one-dimensional Poisson equation $-u'' = f$ with $u(0) = u(1) = 0$. In operator form we write $\mathcal{L}u = f$, with $\mathcal{L} = -d^2/dx^2$.

Our test displacement is built from three vibration modes, $u(x) = 0.8 \sin(\pi x) + 0.4 \sin(2\pi x) + 0.2 \sin(3\pi x)$. The load that produces it is $f = -u'' = \pi^2(0.8 \sin(\pi x) + 1.6 \sin(2\pi x) + 1.8 \sin(3\pi x))$, which peaks at 36.56.

Whether we solve for the displacement, learn the map from load to displacement, or infer the load from measurements, we must describe f and u with a finite list of numbers. How do we choose this list, and how accurate is the representation?

A finite list of numbers and a rule that rebuilds the function

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



How well can finitely many numbers stand in for a function, and what decides the accuracy?

A computer cannot store a curve. Every representation in this course therefore stores a short list of numbers and a rule that turns the list back into a curve. The readings at stations joined by straight lines are one such pair, and the weights on three sine modes are another.

How well can finitely many numbers stand in for a function, and what decides the accuracy?

Outline

- 1 Pointwise sampling
- 2 Functions as vectors
- 3 Norms and the meaning of error
- 4 Function spaces, L^2 , H^1 , and Hilbert spaces
- 5 Inner products, projection, and orthogonality
- 6 Changing the basis
- 7 Convergence, density, and completeness
- 8 Kernels and point evaluation
- 9 Noisy observations, ridge regression, and Gaussian conditioning
- 10 Approximating a family of functions
- 11 Adaptive representations
- 12 Choosing a representation for the cooling fin
- 13 From functions to fields

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

Noise

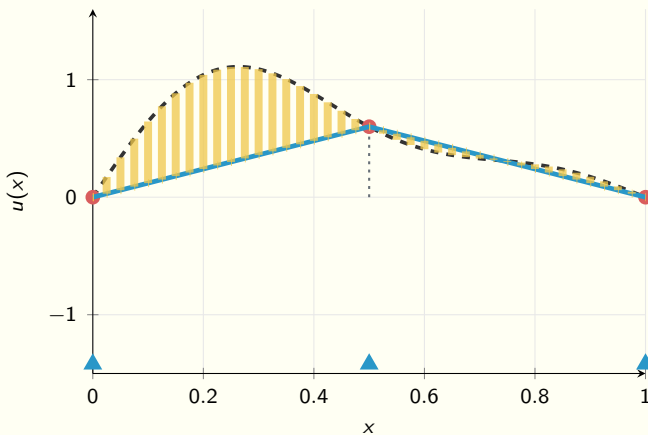
Families

Adaptive

The fin

Fields

Sampling the displacement at fixed stations



three readings
0, 0.60, 0

worst miss, three sensors
0.82, or 73.5 percent

[Try it live](#)

The whole curve is its displacement, and we want to store it in a computer with a handful of numbers. How?

The obvious move is to measure it. Put three sensors along the string, at $x = 0, \frac{1}{2}, 1$.

They give us three readings, 0, 0.60, 0. Everything between the sensors is unknown.

So we guess. The simplest guess connects neighboring readings with straight lines, and we have a curve again.

Now look where the guess is wrong. The lines never rise above 0.60, while the string peaks at 1.11 near $x = 0.23$, so the worst miss is 0.82, about three quarters of the largest displacement.

Would more sensors fix this?

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

Noise

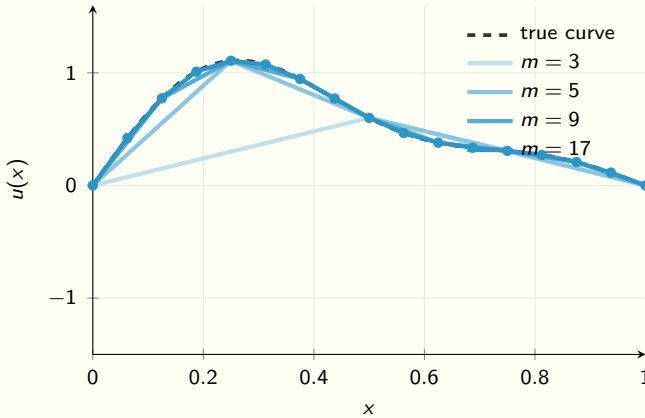
Families

Adaptive

The fin

Fields

Convergence of piecewise-linear interpolation with the spacing



$$m = 3, h = 1/2$$

73.5 percent

$$m = 5, h = 1/4$$

20.1 percent, a factor of 3.7

$$m = 9, h = 1/8$$

6.3 percent, a factor of 3.2

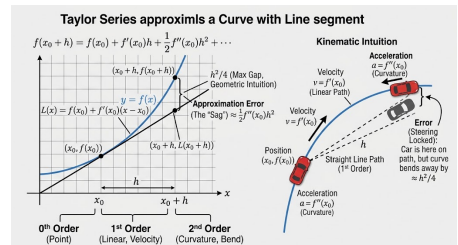
$$m = 17, h = 1/16$$

1.6 percent, a factor of 4.0

So we add sensors. Before adding any, guess how fast the miss should fall. Between two sensors a distance h apart, the curve bends away from a straight line by about $u''h^2/8$ at the middle, so halving h should cut the miss to a quarter. The Taylor series gives the estimate. Along the tangent line at x_0 the curve pulls away by $\frac{1}{2}u''h^2$ after a step h , and the chord through two readings misses by $\frac{1}{2}u''(x - x_i)(x_{i+1} - x)$, which is largest at the midpoint, $u''h^2/8$.

Five sensors give 20.1 percent and nine give 6.3, factors of 3.7 and 3.2, short of four. A segment still spans a large part of the shortest wavelength in the curve, so the bend inside it is far from a parabola and the estimate $u''h^2/8$ is too crude.

Seventeen sensors give 1.6 percent, a factor of 4.0, and from there on every halving divides the miss by four. We can draw these gaps only because we know the curve. What else do the readings alone hide?



Aliasing of a wave that vanishes at every station

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

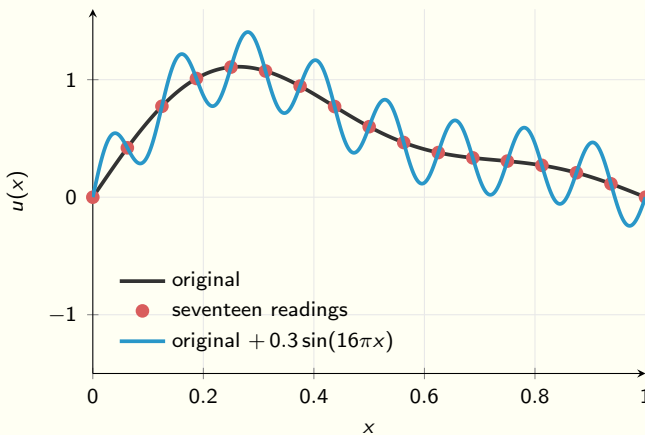
Noise

Families

Adaptive

The fin

Fields



change in the seventeen readings

0

change at each crest
0.3, about a quarter of the largest displacement

[Try it live](#)

Suppose all we have is these seventeen readings. Which curves could have produced them? Take one. Add the wave $0.3 \sin(16\pi x)$ to the displacement. At every station $16\pi x_i$ is a multiple of π , so the wave is zero there and not one reading changes, while the curve moves by 0.3 at each crest, about a quarter of the largest displacement. Two functions that agree at every station are aliases on that grid, and the readings cannot tell them apart. Could a few numbers describe the whole curve at once, instead of one place each?

Grid dependence of sampled representations

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

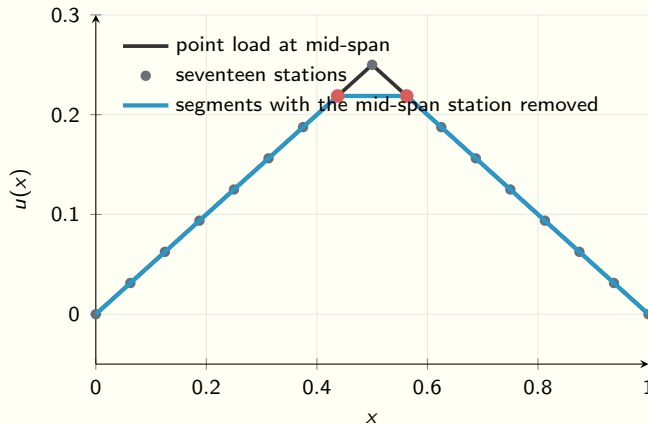
Noise

Families

Adaptive

The fin

Fields



corner, mid-span sensor removed

worst miss 0 to 12.5 percent

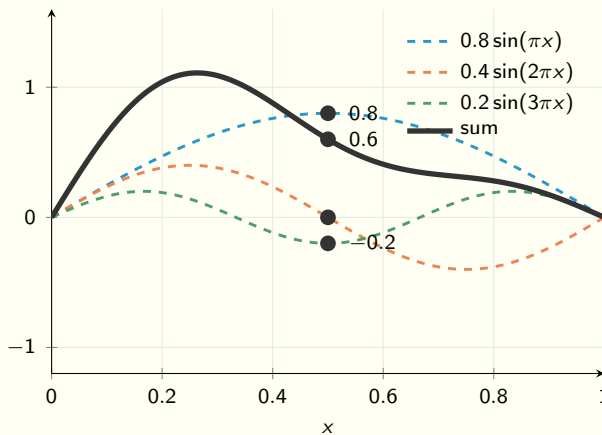
smooth displacement, same sensor removed

1.6 to 1.7 percent

A reading also means something only together with the place it was taken. Two lists from different grids cannot be compared entry by entry, and a sensor that fails leaves a hole for the rule to fill.

Press on the string at mid-span, so it bends into two straight segments with a corner. Take seventeen stations and remove the one on the corner. The straight lines between $x = 0.4375$ and 0.5625 cut the peak off at 0.21875 , so the worst miss jumps from zero to 0.03125 , which is 12.5 percent of the peak 0.25 . Remove the same sensor from the smooth displacement and the miss barely moves, from 1.6 to 1.7 percent. Is there a description in which no number belongs to one place?

Representing a function with coefficients on sine modes



the whole curve

three numbers,
(0.8, 0.4, 0.2)

check at mid-span

$$u\left(\frac{1}{2}\right) = 0.8 - 0.2 = 0.6$$

[Try it live](#)

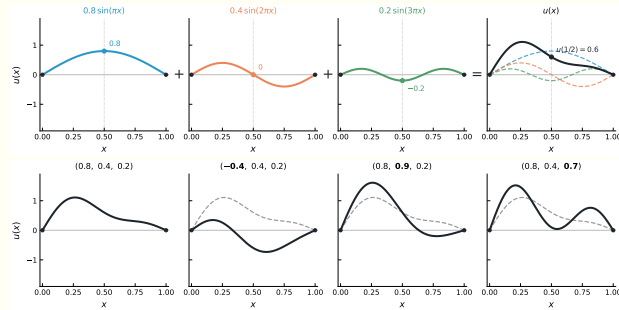
Here is a different idea. Instead of asking where the string is, ask what it is made of. The string vibrates in sine modes, and the plot shows the first one weighted by 0.8.

Add the second mode, weighted by 0.4. Add the third, weighted by 0.2.

Add the three together and the displacement of the string appears. Three numbers, (0.8, 0.4, 0.2), give the whole curve, with no grid anywhere.

Check it at mid-span. The second mode vanishes there, the first and third read 1 and -1 , so $u\left(\frac{1}{2}\right) = 0.8 - 0.2 = 0.6$. And since every sine vanishes at both ends, every curve we can build this way is pinned.

A three-parameter family of string displacements



the family

$$u = a_1 \sin(\pi x) + a_2 \sin(2\pi x) + a_3 \sin(3\pi x)$$

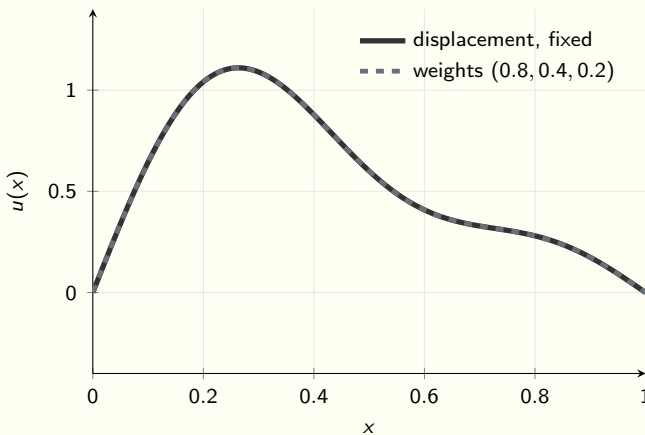
one coefficient changes per panel

$(-0.4, 0.4, 0.2),$
 $(0.8, 0.9, 0.2),$
 $(0.8, 0.4, 0.7)$

The top row shows the weighted modes $0.8 \sin(\pi x)$, $0.4 \sin(2\pi x)$, and $0.2 \sin(3\pi x)$ adding point by point to the displacement, with the three modes repeated as dashed curves beneath their sum in the last panel. The dots are the values at mid-span, 0.8, 0, and -0.2 for the weighted modes and 0.6 for their sum.

The bottom row changes one coefficient per panel while the original shape $(0.8, 0.4, 0.2)$ stays as the dashed curve. Changing a_1 to -0.4 flips the broad hump, changing a_2 to 0.9 tilts the curve toward the left, and changing a_3 to 0.7 adds a third wiggle, and each change moves the whole curve. The three coefficients are coordinates in a three-dimensional space of displacements.

Matching the displacement with three weights



sup, L^2 , H^1 errors
0, 0, 0

solve,
 $a_k = 2 \int_0^1 u \sin(k\pi x) dx$
(0.800, 0.400, 0.200)

[Try it live](#)

Now turn it around. Keep the displacement fixed, start from the weights (0.5, 0, 0), and ask which weights reproduce it. The gray sum misses by 68.5 percent of the largest displacement, and its distance to the displacement is 0.381.

Move the first two weights to 0.8 and 0.4 and the miss falls to 18.0 percent, while the slope error, 46.9 percent, stays the largest of the three.

Add the third weight, 0.2, and all three errors are zero. We need not search. The modes are orthogonal and each has squared norm $\frac{1}{2}$, so each weight is one inner product, $a_k = 2 \int_0^1 u(x) \sin(k\pi x) dx$, which gives (0.800, 0.400, 0.200).

The coefficient list as coordinates in a vector space

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

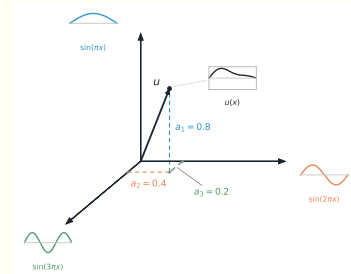
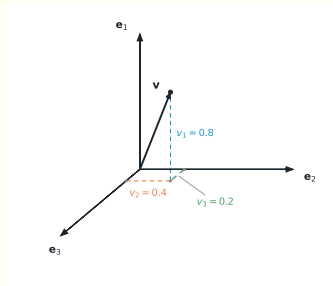
Noise

Families

Adaptive

The fin

Fields

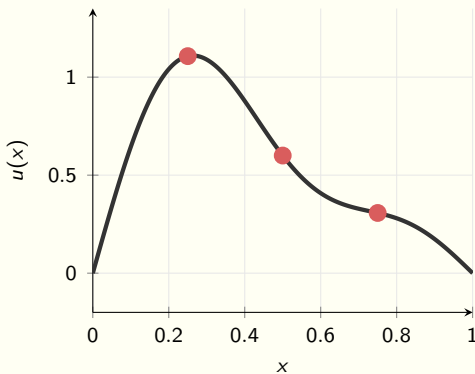


$$u + v = \sum_{i=1}^3 (a_i + b_i) \phi_i, \quad \alpha u = \sum_{i=1}^3 (\alpha a_i) \phi_i$$

Look at what the three numbers are. On the left, $(0.8, 0.4, 0.2)$ is an arrow with three components. On the right, the same three numbers are the displacement.

Add two displacements and their coefficients add, scale one and its coefficients scale, so the family is closed, the three modes are a basis, and a displacement is a point in a three-dimensional space of functions.

Recovering the coefficients from three readings



$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1 & 1/\sqrt{2} \\ 1 & 0 & -1 \\ 1/\sqrt{2} & -1 & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

mid-span alone, $y_2 = u(\frac{1}{2})$
 $a_1 - a_3 = 0.6$, **nothing**
about a_2

quarter spans, y_1, y_3
1.107 and 0.307

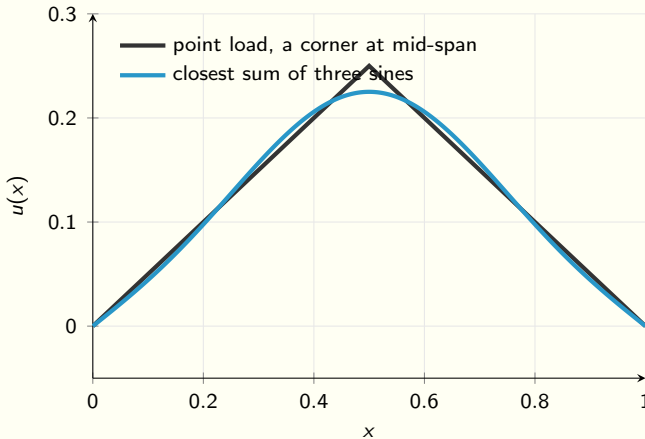
solved
 $a_2 = \frac{y_1 - y_3}{2} = 0.4$,
 $a_1 + a_3 = \frac{y_1 + y_3}{\sqrt{2}} = 1$,
so (0.8, 0.4, 0.2)

Knowing the coefficients and recovering them from measurements are different tasks. A reading at mid-span gives $a_1 - a_3 = 0.6$ and supplies no information about a_2 , since the second mode vanishes there.

The quarter-span readings supply what the mid-span reading misses. At $x = 1/4$ and $x = 3/4$ the second mode takes the values 1 and -1 , while the first and third modes both take the value $1/\sqrt{2}$. The readings are 1.107 and 0.307.

Subtracting them gives $a_2 = (y_1 - y_3)/2 = 0.4$, adding them gives $a_1 + a_3 = (y_1 + y_3)/\sqrt{2} = 1$, and the mid-span reading gives $a_1 - a_3 = 0.6$, which recovers (0.8, 0.4, 0.2). Recovery requires that distinct coefficient lists produce distinct readings, and this matrix has independent columns, orthogonal in fact, with condition number one. The endpoint readings give no equations, because every mode vanishes there, and the aliasing example showed what happens when a mode vanishes at every measurement location.

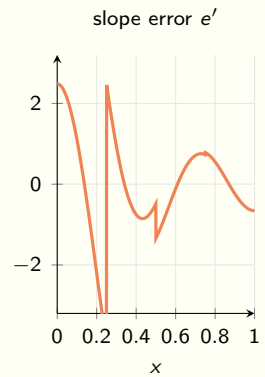
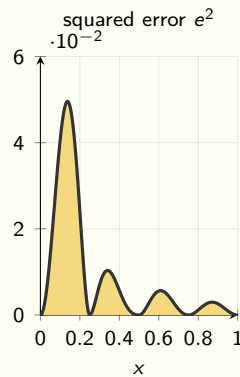
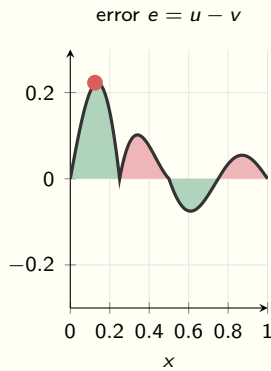
Approximating a function outside the span



So far every curve was smooth, because every sine is smooth. Now press on the string at one point. The point load bends it into two straight segments with a corner at mid-span.

No sum of three sines has a corner, so the best we can do is the closest sum, and it misses by 4.8 percent in the root-mean-square sense. A fixed family represents its own members without error and approximates everything outside it, and the sensors did the same. Before we can say which sum is closest, we have to say how we measure an error.

Measuring error in the sup, L^2 , and H^1 norms



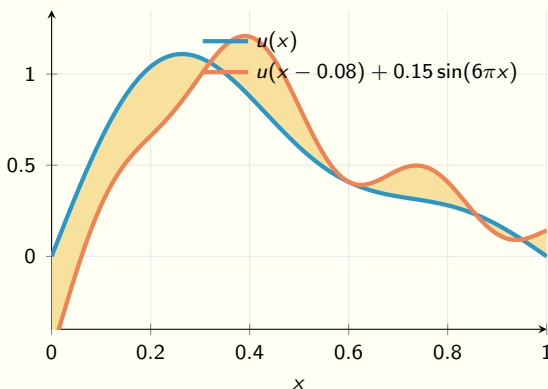
[Try it live](#)

The error of the five-sensor guess is $e = u - v$. How large is it? That depends on what we care about. If it is clearance under the string, only the worst gap matters, 0.22, and that is the sup norm.

If it is the overall misfit, all the gaps count at once. Their average fails, because positive and negative gaps cancel, so square first, integrate, and take the root. That is the L^2 norm, 0.093.

If it is the elastic energy, the values are the wrong thing to look at, since the energy is half the integral of the squared slope. The root of the integrated squared slope error is the H^1 seminorm, 1.20, and half its square is the energy of the error, 0.72. In relative terms the values are off by 14 percent and the slopes by 42.

The L^2 distance



$$\|u-g\|_{L^2} = \left(\int_0^1 (u-g)^2 dx \right)^{1/2}$$

L^2 distance

0.246

signed integral of the gap

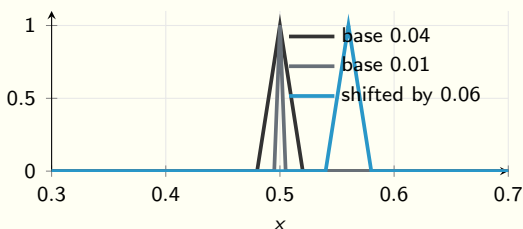
0.0275

[norms notebook](#)

Here is the displacement and a copy of it slid 0.08 to the right, with a small sixth mode added. How far apart are the two curves? The plain integral of the gap is the wrong measure, since positive and negative gaps cancel, and it gives only 0.0275.

Square every vertical gap, the yellow area, integrate, and take the root. That is the L^2 distance, 0.246, and a small slide produces a large distance, the same fact the chapter measures with the shifted spike. The norms notebook has sliders for the shift and for the number of measurement points, and more points approach the same integral.

Disagreement between the sup, L^2 , and H^1 norms



spike, base 0.04
**sup 1, L^2 0.115,
energy 50**

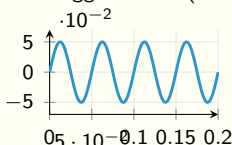
spike, base 0.01
**sup 1, L^2 0.058,
energy 200**

L^2 distance, spike to
shifted spike
 $\sqrt{2} \times 0.115 = 0.163$

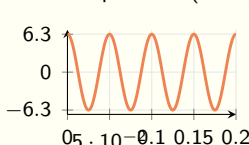
wiggle $0.05 \sin(40\pi x)$
**sup 0.05, L^2 0.035,
 H^1 4.44**

[Try it live](#)

the wiggle $0.05 \sin(40\pi x)$



its slope $2\pi \cos(40\pi x)$



Can the three disagree by a lot? Take a spike of height one on a base of width 0.04. Its sup norm is one. Its L^2 norm is $\sqrt{b/3} = 0.115$, and its slopes are $\pm 2/b$, so its elastic energy is $2/b = 50$.

Quarter the base. The sup norm stays at one, the L^2 norm halves to 0.058, and the energy climbs to 200. No multiple of the L^2 norm bounds the sup norm, and a displacement of finite energy cannot contain a spike of zero width.

Shift the spike by one and a half base widths, so it overlaps the original nowhere. The L^2 distance between the two is $\sqrt{2} \times 0.115 = 0.163$, because the L^2 distance compares the two point by point and they never overlap. A small fast wiggle, $0.05 \sin(40\pi x)$, is small in the sup and L^2 norms, 0.05 and 0.035, and its slope swings to ± 6.3 , so its H^1 seminorm is 4.44.

Sampling

Vectors

Bases

Spaces

Projection

Bases

Convergence

Kernels

Noise

Families

Adaptive

The fin

Fields

Definition of a norm and a seminorm

Norm

$$\|u\| = 0 \text{ only for } u = 0, \quad \|\alpha u\| = |\alpha| \|u\|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

Seminorm

$$|c|_{H^1} = \left(\int_0^1 (c')^2 dx \right)^{1/2} = 0 \quad \text{for every constant } c, \quad \text{and on the pinned string}$$

To compute the “closest” approximation of a mathematical function, you must first define the mathematical ruler, or “norm,” used to measure distance and error.

The Three Rules of a True Norm

A valid measurement norm must satisfy three conditions:

- **Definiteness:** $\|u\| = 0 \iff u = 0$. A function measures a size of zero if and only if it is completely flat at zero.
- **Absolute Homogeneity:** $\|\alpha u\| = |\alpha| \|u\|$. Scaling a function scales its measured size by the exact same absolute factor.
- **Triangle Inequality:** $\|u + v\| \leq \|u\| + \|v\|$. The direct path between points is always shorter than or equal to an indirect combined path.

The Seminorm Failure

Evaluating a function’s size purely by its slope creates a mathematical contradiction. An elevated flat line (e.g., $y = 5$) has a slope of zero, meaning the formula would calculate a size of zero for a non-zero function. Because it fails the definiteness rule, this measurement is downgraded to a “seminorm.”

The Pinned String Loophole

Applying physical boundary conditions fixes the seminorm failure. If a string is pinned at both ends ($x = 0$ and $x = 1$), it is impossible for it to form an elevated flat line. The only possible constant state is resting exactly at zero. This physical constraint eliminates rule-breaking constants, satisfying the definiteness requirement and upgrading the slope rule back to a true norm.

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

Noise

Families

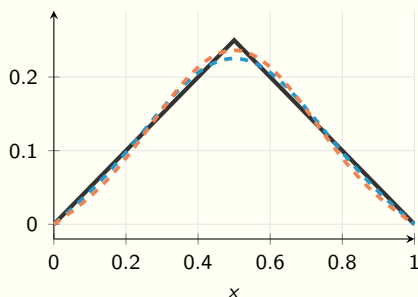
Adaptive

The fin

Fields

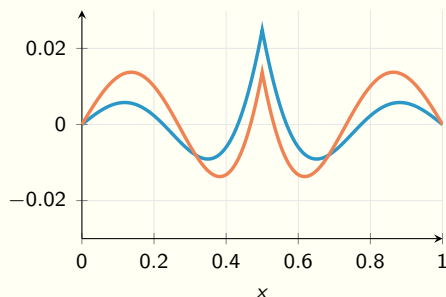
Closest three-sine sums to the corner in the L^2 and sup norms

the corner (black), closest in L^2 (blue) and in sup (orange)



L^2 -closest,
(0.2026, 0, -0.0225)
**sup 9.9, L^2 4.8, H^1
31.5 percent**

the gap $u - v$



sup-closest,
(0.2045, 0, -0.0318)
**sup 5.5, L^2 6.7, H^1
33.9 percent**

Now go back to the corner and ask which sum of three sines is closest. In L^2 the closest sum has weights $c_k = 2 \sin(k\pi/2)/(k\pi)^2$, that is (0.2026, 0, -0.0225), and it is also the closest in H^1 , because the sines are orthogonal in both inner products. It misses by 9.9 percent of the peak in the largest gap, 4.8 percent in the root-mean-square gap, and 31.5 percent in the slopes, and its largest gap sits at the corner.

The sum closest in the sup norm has different weights, (0.2045, 0, -0.0318), found by minimizing the largest gap numerically. Its gap reaches the same size at several points, and it trades the other way, 5.5 percent in the largest gap against 6.7 in the root-mean-square gap and 33.9 in the slopes. Each sum is closer in the norm it minimizes, and no sum of three sines reaches the corner itself. How do we find the L^2 -closest sum without searching?

The space L^2 of square-integrable functions

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

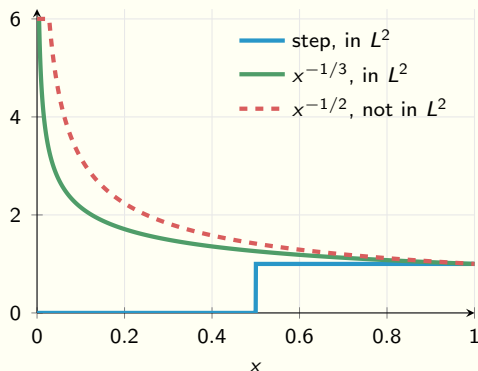
Noise

Families

Adaptive

The fin

Fields



$$\|f\|_{L^2}^2 = \int_0^1 f(x)^2 dx < \infty$$

$$\langle f, g \rangle = \int_0^1 fg dx$$

$$\int_0^1 (x^{-1/3})^2 dx$$

3, finite although
 $x^{-1/3}$ is unbounded

$$\int_0^1 (x^{-1/2})^2 dx = \int_0^1 x^{-1} dx$$

diverges

two functions equal except
at one point

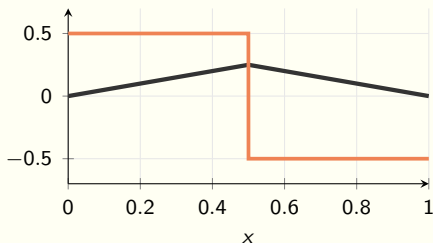
L^2 distance 0

Every norm so far measured a function on the string, and each one needs its own space of functions for which it is finite. The space $L^2(0, 1)$ consists of the functions whose square has a finite integral, and its inner product is $\int fg dx$, the signed overlap the projection section uses.

L^2 does not care about roughness. A unit step is in L^2 , and so is $x^{-1/3}$, which grows without bound near zero, because $\int_0^1 x^{-2/3} dx = 3$ is finite. The function $x^{-1/2}$ is not, since its square $1/x$ has an integral that diverges. Two functions that differ at one point have L^2 distance zero, so an element of L^2 is a class of functions that agree except on a set of zero length, and it has no value at a single point. What does a space need to know about slopes?

The space H^1 of finite-energy functions

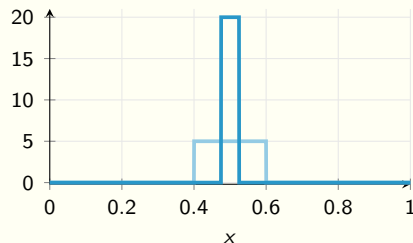
the corner $\min(x, 1-x)/2$ and its slope, in H_0^1



$$\|u\|_{H^1}^2 = \int_0^1 u^2 dx + \int_0^1 (u')^2 dx,$$

ramp of width w ,
 $\int (u')^2 dx$
 $1/w$, so 5 at $w = 0.2$
and 20 at $w = 0.05$

ramps toward a step, and their slopes $1/w$



$$\langle u, v \rangle_{H^1} = \int_0^1 (uv + u'v') dx$$

in one dimension

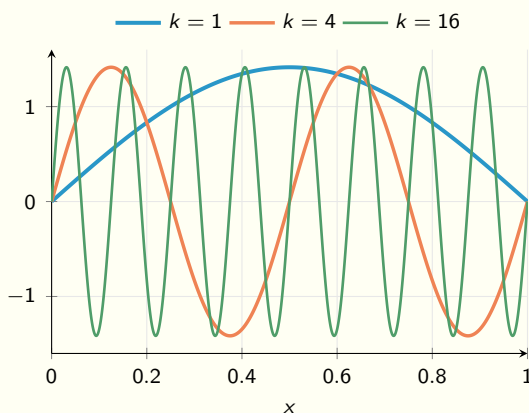
$$|u(y) - u(x)| \leq \sqrt{|y - x|} \|u'\|_{L^2}$$

The Sobolev space $H^1(0, 1)$ consists of the functions in L^2 whose first derivative, in the weak sense, is also in L^2 . Its norm adds the slope term, $\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|u'\|_{L^2}^2$. For the pinned string we use H_0^1 , the functions of H^1 that vanish at both ends, where the slope term alone is a norm and half its square is the elastic energy.

The point-load corner is in H_0^1 , because its slope is a step, $\pm \frac{1}{2}$, and a step is square integrable. A step is not in H^1 . Ramps of width w approach it, and their slope $1/w$ has squared integral $1/w$, which is 5 at $w = 0.2$ and 20 at $w = 0.05$ and grows without bound as w shrinks.

The slope bound has a consequence in one dimension. For every u in H^1 , $|u(y) - u(x)| \leq \sqrt{|y - x|} \|u'\|_{L^2}$, so values at nearby points are close and every element has a continuous version. In two and more dimensions this fails, and H^1 functions can be unbounded at isolated points.

Equal L^2 size, growing H^1 size



$$\phi_k(x) = \sqrt{2} \sin(k\pi x)$$

$$\|\phi_k\|_{L^2} = 1, \quad \|\phi_k\|_{H^1} = k\pi$$

$$\|\phi_k\|_{H^1} \text{ at } k=1$$

$$3.14$$

$$k=4$$

$$12.57$$

$$k=16$$

$$50.27$$

distance between two
different modes

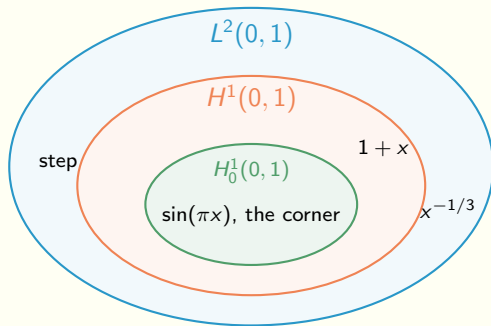
$$\sqrt{2} = 1.414$$

Here are the unit modes $\phi_k = \sqrt{2} \sin(k\pi x)$ for $k = 1, 4$, and 16 . Every one has L^2 norm one, so L^2 gives the fast wiggle the same size as the broad hump. The slope is $\sqrt{2} k\pi \cos(k\pi x)$, so the H^1 seminorm is $k\pi$, 3.14, 12.57, and 50.27. The H^1 size grows in proportion to the frequency, and the L^2 size stays at one.

The modes are also orthonormal, so any two of them are $\sqrt{2} = 1.414$ apart in L^2 . An infinite list of unit vectors that stay $\sqrt{2}$ apart has no convergent subsequence, and that fact is the next frame's subject.

Hilbert spaces and the nesting $H_0^1 \subset H^1 \subset L^2$

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



Hilbert space

A vector space with an inner product in which every Cauchy sequence converges to a member of the space.

Hilbert spaces

$\mathbb{R}^n$, $L^2(0,1)$, $H^1(0,1)$,
 $H_0^1(0,1)$

not complete

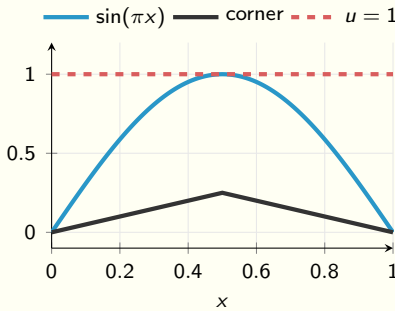
**continuous functions
with the L^2 distance**

The three spaces nest. Every function of H_0^1 is in H^1 , and every function of H^1 is in L^2 , so $H_0^1 \subset H^1 \subset L^2$. The pinned sine and the point-load corner are in H_0^1 , the line $1 + x$ is in H^1 but does not vanish at the ends, and the step and $x^{-1/3}$ are in L^2 only.

A Hilbert space is a vector space with an inner product that is also complete, so every Cauchy sequence, a sequence whose members eventually come as close together as we like, has its limit inside the space. $\mathbb{R}^n$ is one, and so are L^2 , H^1 , and H_0^1 with their own inner products. The continuous functions with the L^2 distance are not, since ramps steepening toward a step form a Cauchy sequence whose limit, the step, is not continuous.

Completeness is what the projection theorem needs for infinitely many shapes. A minimizing sequence of better and better approximations then has its limit in the space, so a closest member exists.

The space H_0^1 and the Poincaré inequality



$\|u\|_{L^2}/|u|_{H^1}$ for $\sin(\pi x)$
 $0.3183 = 1/\pi$,
equality

for the corner
 $0.1443/0.5 = 0.2887$

$u = 1$, in H^1 only
 $|u|_{H^1} = 0$, **the bound fails**

The space $H_0^1(0, 1)$ consists of the functions of H^1 that vanish at both ends. In one dimension every H^1 function has a continuous version, so the end values $u(0)$ and $u(1)$ make sense, and the map $\gamma u = (u(0), u(1))$ is the trace. H_0^1 is the kernel of the trace, and it is also the closure in the H^1 norm of the smooth functions that vanish near the ends. On a domain Ω in two or three dimensions an H^1 function need not be continuous, but the trace still extends to a bounded map on $H^1(\Omega)$, and $H_0^1(\Omega)$ is again its kernel.

Pinning the ends controls the size of u by its slope. Write $u = \sum c_k \sqrt{2} \sin(k\pi x)$, so $\|u\|_{L^2}^2 = \sum c_k^2$ and $|u|_{H^1}^2 = \sum (k\pi)^2 c_k^2 \geq \pi^2 \sum c_k^2$. That comparison is the Poincaré inequality $\|u\|_{L^2} \leq |u|_{H^1}/\pi$, and the lowest mode $\sin(\pi x)$ attains it. The corner has ratio $0.1443/0.5 = 0.2887$ and the three-sine displacement has 0.2278 , both below $1/\pi = 0.3183$. The constant $u = 1$ is in H^1 with zero slope and L^2 norm one, so without the pinned ends no such bound holds.

On H_0^1 the slope term alone is therefore a norm, equivalent to the full H^1 norm, since $\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + |u|_{H^1}^2 \leq (1 + 1/\pi^2) |u|_{H^1}^2$. Half of $|u|_{H^1}^2$ is the elastic energy of the string, so the energy norm and the H^1 norm measure the same convergence. On a bounded domain Ω the constant is $1/\sqrt{\lambda_1}$, where λ_1 is the lowest eigenvalue of $-\Delta$ with zero boundary values, and $\lambda_1 = \pi^2$ on $(0, 1)$.

The spaces L^2 , H^1 , and H_0^1 compared

- Sampling
- Vectors
- Norms
- Spaces
- Projection
- Bases
- Convergence
- Kernels
- Noise
- Families
- Adaptive
- The fin
- Fields

space	$u \in L^2$	$\nabla u \in L^2$	inner product	boundary values	role in $-\Delta u = f$
$L^2(\Omega)$	yes	not required	$\int_{\Omega} uv \, dx$	none, no values at points	the load f and data misfit
$H^1(\Omega)$	yes	yes	$\int_{\Omega} (uv + \nabla u \cdot \nabla v) \, dx$	trace exists, none imposed	free (Neumann) boundaries, through the domain
$H_0^1(\Omega)$	yes	yes	$\int_{\Omega} \nabla u \cdot \nabla v \, dx$	$u = 0$ on $\partial\Omega$	trial and test functions, fixed (Dirichlet) boundaries

The table lists the three spaces side by side on a domain Ω , with $\nabla u = u'$ on the string. L^2 asks only that u^2 has a finite integral, so its members have no values at points and no boundary values, and it is the space of loads and of data misfits. H^1 adds a square-integrable gradient, which gives every member a trace on the boundary, but it imposes no condition there. A free end, where the weak form supplies the condition $u' = 0$ or a given flux, uses H^1 . H_0^1 sets the trace to zero, which is the pinned string and the clamped membrane, and it is the space in which Galerkin with hats looks for the displacement and takes its test functions.

The inner products follow the same order. L^2 uses $\int uv \, dx$, H^1 adds the gradient term, and H_0^1 can drop the $\int uv \, dx$ term because the Poincaré inequality makes the gradient term alone a norm there.

Compactness from bounded energy

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields

$$u = \sum_{k \geq 1} c_k \sqrt{2} \sin(k\pi x), \quad |u|_{H^1}^2 = \sum_{k \geq 1} (k\pi)^2 c_k^2$$

$$|u|_{H^1} \leq M \implies \left(\sum_{k > K} c_k^2 \right)^{1/2} \leq \frac{M}{(K+1)\pi}$$

unit ball of L^2
**the modes stay $\sqrt{2}$
apart, no
subsequence
converges**

$M = 10$, tail beyond $K = 4$
at most 0.637

$M = 10$, tail beyond
 $K = 16$
at most 0.187

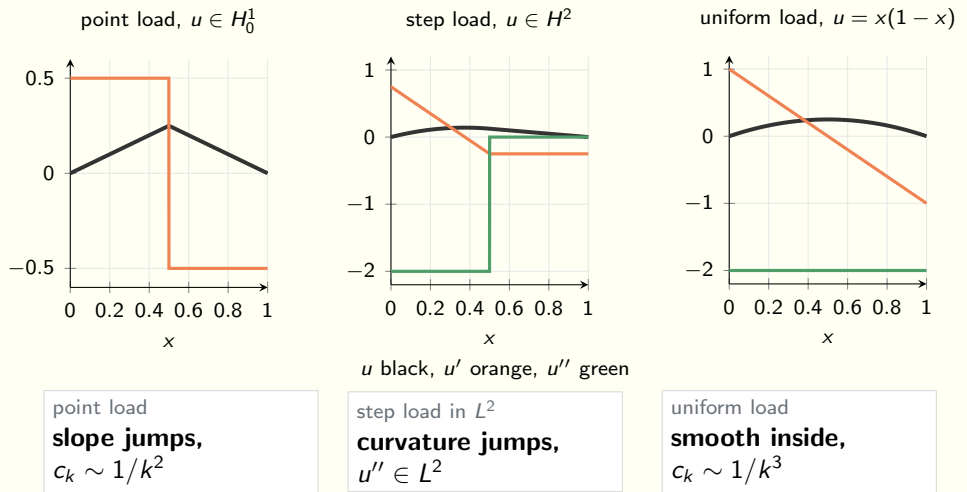
Rellich, on a bounded
interval
**a bounded set of H_0^1
has an L^2 -convergent
subsequence**

In $\mathbb{R}^n$ every bounded sequence has a convergent subsequence. In L^2 this fails. The unit modes $\sqrt{2} \sin(k\pi x)$ all have norm one and stay $\sqrt{2}$ apart, so no subsequence converges, and the unit ball of L^2 is not compact.

A bound on the energy repairs it. Write u in the sine modes, so $|u|_{H^1}^2 = \sum (k\pi)^2 c_k^2$. If $|u|_{H^1} \leq M$, the coefficients beyond mode K satisfy $\sum_{k > K} c_k^2 \leq M^2 / ((K+1)\pi)^2$, so the tail has L^2 norm at most $M / ((K+1)\pi)$, 0.637 beyond mode 4 and 0.187 beyond mode 16 when $M = 10$. Bounded energy leaves no room for fast wiggles, so every such function is close to one with finitely many modes.

This is Rellich's theorem on a bounded interval, a bounded set of H_0^1 has a subsequence that converges in L^2 . It is why truncating to finitely many modes or hats has an error bound when the energy is bounded, and why minimizers of energy functionals exist.

Regularity from the load to the displacement



Regularity is how many derivatives a function has in a given space, and for the string it passes from the load to the displacement. Since $-u'' = f$, the displacement has two more derivatives than the load.

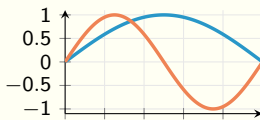
A point load at mid-span is not a function in L^2 , and the displacement is the corner, in H_0^1 with a slope that jumps. A step load, 2 on the left half and 0 on the right, is in L^2 , so u'' is in L^2 and u is in H^2 , with a continuous slope and a curvature that jumps at mid-span, and its peak is 0.1405 at $x = 0.375$. A uniform load gives the parabola $x(1 - x)$, smooth inside the string.

The sine coefficients read the same regularity. Each derivative the odd continuation keeps continuous adds one power of k to the decay, $1/k$ for the uniform load itself, $1/k^2$ for the corner, and $1/k^3$ for the parabola. The weak form of $-u'' = f$ needs only one derivative on u , which is why H_0^1 is the natural space for the string and why finite elements with hats, which have one square-integrable derivative, solve it.

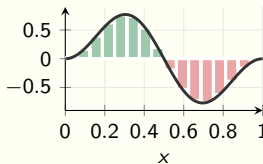
The inner product as signed overlap

$f = \sin(\pi x)$ blue, g orange, and below them fg , green where positive and red where negative

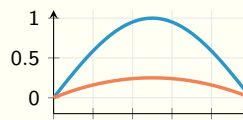
$g = \sin(2\pi x)$



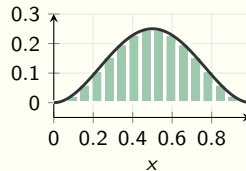
fg , net area 0



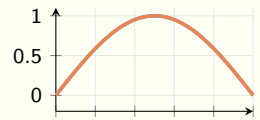
$g = x(1 - x)$



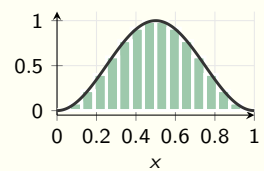
fg , net area 0.129



$g = f$



fg , net area 0.500



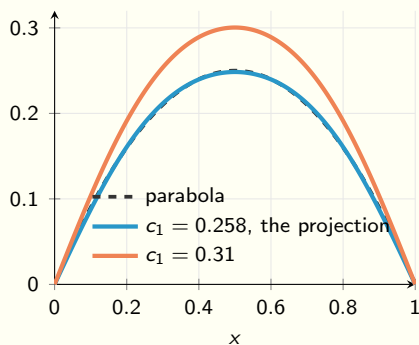
$$\langle f, g \rangle = \int_0^1 f(x) g(x) dx \approx \frac{1}{16} \sum_{i=1}^{16} f(x_i) g(x_i)$$

The uniform load $f = 2$ bends the string into the parabola $x(1 - x)$, and no sum of three sines equals it. In ordinary geometry a perpendicular dropped from the point onto the plane finds the closest point, and a perpendicular needs a dot product. For two vectors the dot product multiplies matching entries and adds. Do the same for two functions sampled at sixteen points, and let the points crowd together, and the sum becomes the integral of the product, $\langle f, g \rangle = \int_0^1 fg dx$. Look at each pair on top, then at its product below. Where f and g have the same sign the product is positive, a green bar, and where they have opposite signs it is negative, a red bar. The inner product is the green area minus the red area, the net signed overlap.

Try $\sin(\pi x)$ against $\sin(2\pi x)$. One is even about mid-span and the other odd, so every green bar on the left has a red twin on the right, the net area is zero, and the modes are orthogonal. Now $\sin(\pi x)$ against the parabola $x(1 - x)$. The two never oppose in sign, so every bar is green and the area is 0.129, which is $4/\pi^3$, and integrating $-u'' = 2$ by parts twice gives it without integrating the parabola at all.

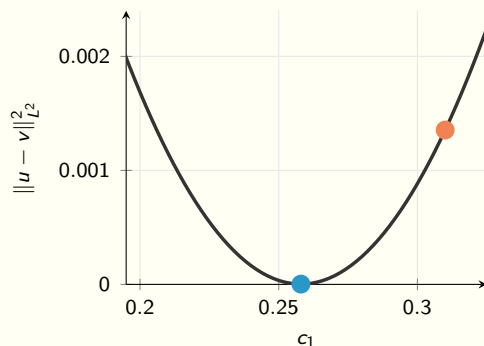
Last, $\sin(\pi x)$ against itself. The product is a square, never negative, and its area 0.5 is the squared length $\|\sin(\pi x)\|^2$, so the inner product of a function with itself is the square of its L^2 norm. The sixteen-sample sums give 0.000, 0.129, and 0.500, the same as the integrals to three digits.

Minimizing squared error via orthogonal projection



$c_1 = 8/\pi^3 = 0.258$
error 0.87 percent

$c_1 = 0.31$, twenty percent
high
**error norm 23 times
larger**



[Try it live](#)

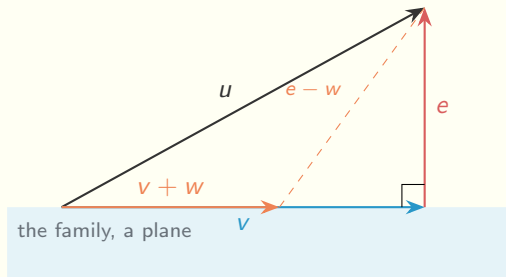
Fix $c_2 = 0$ and $c_3 = 8/(27\pi^3)$, and slide c_1 . The squared error is a quadratic in c_1 , and its slope is $-2\langle u - v, \sin(\pi x) \rangle$.

The bottom of the bowl sits where that inner product vanishes, where the error is orthogonal to the mode. That gives $c_1 = 2\langle u, \sin(\pi x) \rangle = 8/\pi^3 = 0.258$, one division per coefficient because the modes are orthogonal to each other, and an error of 0.87 percent.

Push c_1 twenty percent too high, to 0.31. To the eye the curve looks as good, and its error norm is 23 times larger, since the squared error rises from 2.5×10^{-6} to 1.35×10^{-3} .

Uniqueness of the projection and the Pythagorean identity

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



$$\|e - w\|^2 = \|e\|^2 + \|w\|^2$$
$$\|u\|^2 = \|v\|^2 + \|e\|^2, \quad 0.033333 = 0.03$$

Why is the perpendicular the best we can do? Write $u = v + e$ with e orthogonal to every mode. Move to any other combination $v + w$. The new error is $e - w$, and because e is orthogonal to w , $\|e - w\|^2 = \|e\|^2 + \|w\|^2$. Every move away from v adds its own squared length to the error, so v is the closest combination and the only one.

Take $w = -v$ and we get Pythagoras, $\|u\|^2 = \|v\|^2 + \|e\|^2$, which predicts the error before we draw anything. $\|u\|^2 = 1/30 = 0.033333$ and $\|v\|^2 = 0.033331$, so $\|e\|^2 = 2.5 \times 10^{-6}$, and the 0.87 percent follows.

The projection theorem

Projection theorem

In a space with an inner product, every finite-dimensional linear subspace contains one closest member to each target, and that member is the one whose error is orthogonal to the subspace.

$$\langle u - v, \phi_j \rangle = 0 \quad \text{for every shape } \phi_j, \quad c_k = \frac{\langle u, \phi_k \rangle}{\langle \phi_k, \phi_k \rangle} \quad \text{when the shapes are orthogonal}$$

Orthogonality gives one linear equation per shape, so the closest member is computed and never searched for, and when the shapes are orthogonal to one another each coefficient is a single division. The Fourier coefficients of a function are these divisions, one per mode.

For infinitely many shapes the theorem needs a closed linear subspace of a Hilbert space. For a nonlinear family, such as networks with trainable breakpoints, a closest member need not be unique, since every point of the unit circle is equally close to the origin. Is the inner product itself fixed, or a choice?

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

Noise

Families

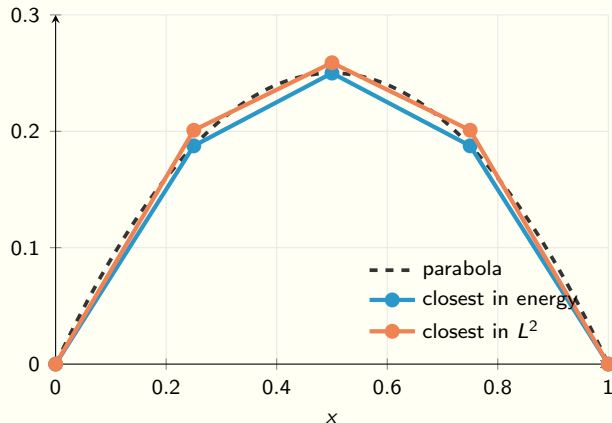
Adaptive

The fin

Fields

Projection in the L^2 and energy inner products

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



closest in energy, heights
0.1875, 0.25, 0.1875

closest in L^2 , heights
0.2009, 0.2589, 0.2009

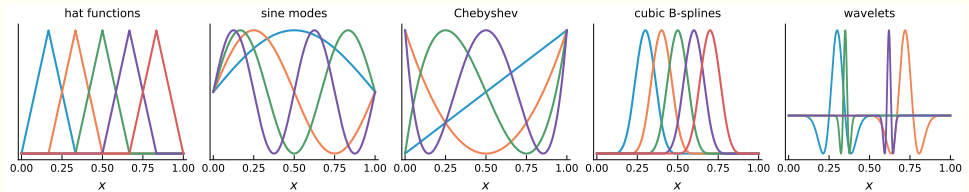
L^2 error, L^2 against energy
3.3 against 6.3 percent

H^1 error, L^2 against energy
26.0 against 25.0 percent

The inner product was our choice. The energy inner product $\int f'g' dx$ is another, and it gives another closest member. Keep the parabola and take a new family, straight segments with breaks at 0.25, 0.5, 0.75. In energy the closest member passes through the parabola's own values, 0.1875, 0.25, 0.1875, and it is the straight-segment guess from the first slides, because its slope error integrates to zero across every segment.

In L^2 the closest member lifts the heights to 0.2009, 0.2589, 0.2009, which opens small gaps at the stations and closes larger ones between them. Each is closer in the norm it minimizes. For the sines the two projections coincide, because the sines are orthogonal in both inner products.

Five fixed families of shapes



local
hats, one per node

global
**sine modes,
Chebyshev
polynomials**

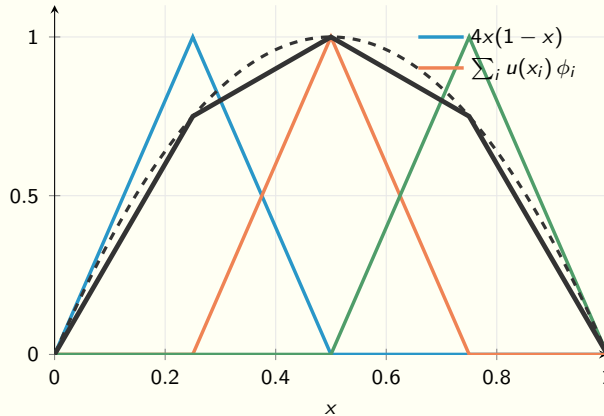
smooth and local
cubic B-splines

local in position and scale
wavelets

The sine modes are one basis among many. Hat functions are local, one per node, and zero outside the two segments beside it. Sine modes are global and are the string's own vibration shapes, and Chebyshev polynomials are global as well. Cubic B-splines are smooth and local at once, and a wavelet is local in position and in scale.

Different bases localize information differently. A global shape changes the curve everywhere when its coefficient changes, and a local shape changes it only near its support. Which basis describes a function with few numbers depends on the function, and the corner shows it next.

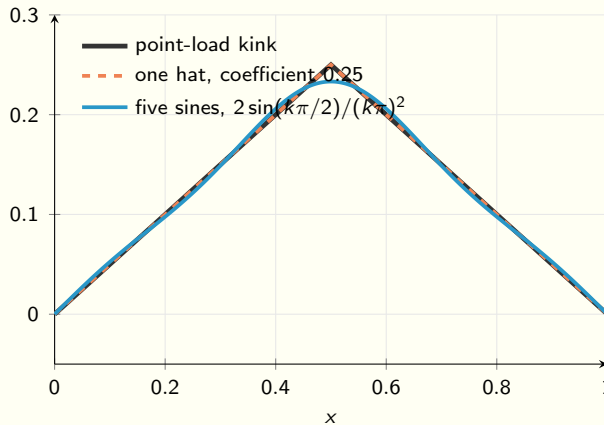
Piecewise-linear interpolation as a hat-function expansion



$$u_h(x) = \sum_i u(x_i) \phi_i(x)$$

Look again at the straight segments. They are a basis too. Take one shape per interior station, equal to one at its own station, zero at the others, and straight in between, a hat function. Weight each hat by the reading at its station and add. The sum is the straight-segment guess, $u_h(x) = \sum_i u(x_i) \phi_i(x)$, so the readings are the coefficients, and the energy projection found this same curve a moment ago.

Representing a kink in the hat basis and the sine basis



one hat on the corner
one number, 0.25, no error

three hats, five stations
0.125, 0.25, 0.125, **no error**

sines, three and thirty-three modes
4.8 and 0.20 percent

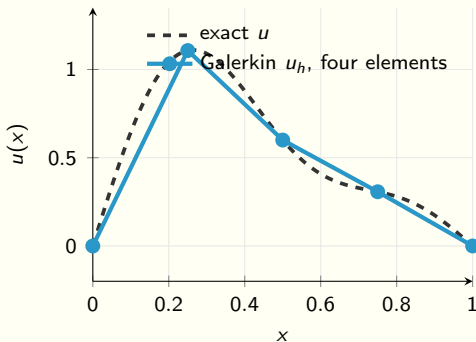
[Try it live](#)

Back to the point load. Which shapes describe its corner with few numbers?

One hat whose node sits on the corner does it with one number, 0.25. On the five-station grid three hats do it with 0.125, 0.25, 0.125 and zero error.

The sines need the infinite list $c_k = 2 \sin(k\pi/2)/(k\pi)^2$, which falls like $1/k^2$ and leaves 4.8 percent after three modes and 0.20 percent after thirty-three. A shape that shares the corner needs one number where smooth shapes need infinitely many.

Galerkin's method with hat functions



$$\int_0^1 u_h' \phi_i' dx = \int_0^1 f \phi_i dx \quad \text{for every hat } \phi_i$$

$$\mathbf{K}\mathbf{u} = \mathbf{F}, \quad K_{ii} = \frac{2}{h}, \quad K_{i,i\pm 1} = -\frac{1}{h}$$

largest nodal deviation, 4
to 16 elements
 4.4×10^{-16}

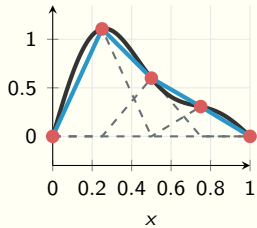
error slopes on log axes, L^2
and energy
2.00 and 1.00

So far we set the hat weights to the readings. Galerkin's method finds them from the equation instead, without readings. A hat has a corner, so the strong form $-u_h'' = f$ cannot be evaluated, and the weak form asks only for one derivative, $\int u_h' v' dx = \int f v dx$. Testing the weak form against each hat in turn gives $\mathbf{K}\mathbf{u} = \mathbf{F}$, with $K_{ii} = 2/h$ and $K_{i,i\pm 1} = -1/h$, because hat derivatives are steps of height $\pm 1/h$ that overlap only between neighbors.

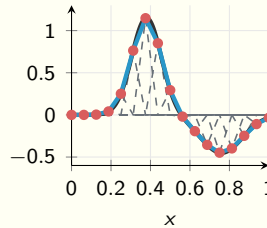
The error is orthogonal to every hat in the energy inner product, so by the projection theorem u_h is the closest member of the hat space in energy. For the string, the nodal values land on the exact displacement, to 4.4×10^{-16} at four, eight, and sixteen elements, because the Green's function of $-d^2/dx^2$ is a tent that lies in the hat space. Courant proposed these hats in 1943, and the finite element method is built on them. Measured over four to two hundred fifty-six elements, the error falls with slope 2.00 in L^2 and 1.00 in energy, since the values are accurate to order h^2 and the staircase slopes only to order h .

Approximating any shape with hat functions

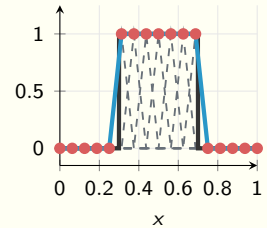
the string, five stations



a bump, seventeen stations



a square pulse, seventeen stations



$$u_h(x) = \sum_i u(x_i) \phi_i(x), \quad \phi_i(x_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

string, largest gap at 5, 17,
33 stations
20.1, 1.6, 0.4 percent

bump, largest gap at 9, 33,
65 stations
13.4, 2.1, 0.5 percent

pulse, largest gap and
RMS gap at 17
80 and 23.3 percent

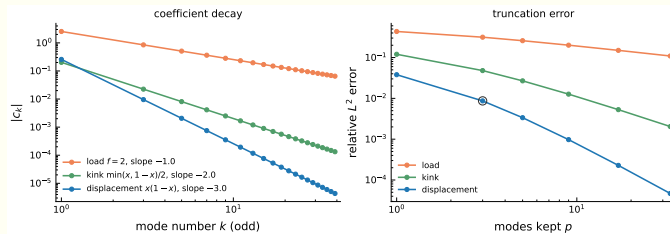
[Try it live](#)

Look again at the straight segments. They are a basis too. Take one shape per station, equal to one at its own station, zero at every other station, and straight in between, a hat function. Weight each hat by the reading at its station and add, and the sum is the straight-segment guess, with the readings as its coefficients. A hat is zero outside the two segments beside its station, so each coefficient changes the curve only there.

The finite element method is built on this basis. Courant proposed piecewise-linear hats in 1943, and Galerkin's method chooses their weights by making the error orthogonal to every hat in the energy inner product, which for the string gives the readings, as the energy projection showed. Hats model every continuous function on the interval, since the gap goes to zero as the spacing h shrinks, and for $|u''| \leq M$ the gap is at most $Mh^2/8$. The string misses by 20.1 percent at five stations, 1.6 at seventeen, and 0.4 at thirty-three. The bump from the coefficient-decay lab is narrower and needs more hats, 13.4 percent at nine stations, 2.1 at thirty-three, and 0.5 at sixty-five.

The square pulse is the case hats cannot finish. Every sum of hats is continuous and the pulse jumps, so the largest gap stays between 60 and 80 percent at every station count. The root-mean-square gap still falls, 23.3 percent at seventeen stations and 11.6 at sixty-five, halving each time the stations are quadrupled. Which shapes to use depends on the function, and so far we have placed them by hand.

Decay rates of sine coefficients



load $f = 2$, odd k
 $|c_k| = 8/(k\pi)$, slope
 -1.0

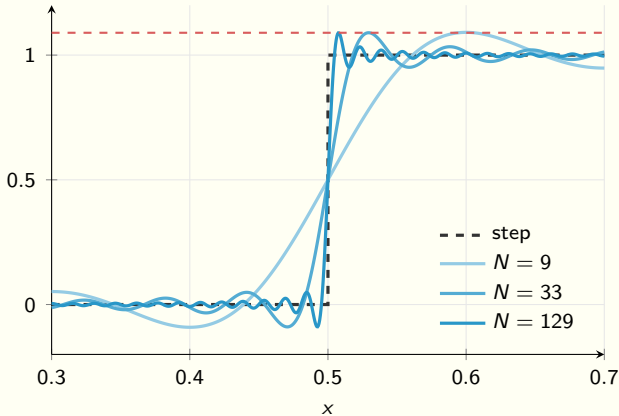
kink, odd k
 $|c_k| = 2/(k\pi)^2$, slope
 -2.0

parabola, odd k
 $|c_k| = 8/(k\pi)^3$, slope
 -3.0

How fast do coefficients fall? Put three functions on the sine modes. The load $f = 2$ has $c_k = 8/(k\pi)$ for odd k and zero for even k , the kink has $c_k = 2\sin(k\pi/2)/(k\pi)^2$, which alternates in sign over odd k , and the parabola has $c_k = 8/(k\pi)^3$ for odd k , since $-u'' = f$ divides the load's coefficients by $(k\pi)^2$. The plot shows the magnitudes of the nonzero coefficients.

A sine series represents the odd continuation of the function, and each derivative that continuation keeps continuous gains one power of k , which is what the slopes -1.0 , -2.0 , -3.0 show. One mode leaves 44 percent of the load and thirty-three still leave 11, because every sine vanishes at the ends where the load does not. What happens beside a jump, where no derivative is continuous?

The Gibbs phenomenon at a jump



overshoot, $N = 9$
0.091

overshoot, $N = 33$
0.090

overshoot, $N = 129$
0.089, **toward** 0.0895

Sum the sine series of a step. With nine modes the sum overshoots the top of the step beside the jump.

Add modes. The oscillation squeezes toward the jump, and the overshoot stays.

The overshoot tends to 0.0895 of the jump and never below it, so the series does not converge in the sup norm. It does converge in L^2 , because the region of large error narrows. That is the Gibbs phenomenon.

Conditioning of the Gram matrix for nearly parallel shapes

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

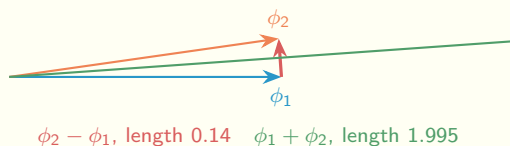
Noise

Families

Adaptive

The fin

Fields



$$\mathbf{G} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \rho = 0.99, \quad \text{cond } \mathbf{G}$$

add $0.1(\phi_1 - \phi_2)$
target moves 0.014,
coefficients 0.1 and
-0.1

add $0.1(\phi_1 + \phi_2)$
target moves 0.2,
fourteen times more

$\sqrt{199}$
14.1

neighboring hats
 $\rho = 1/4$

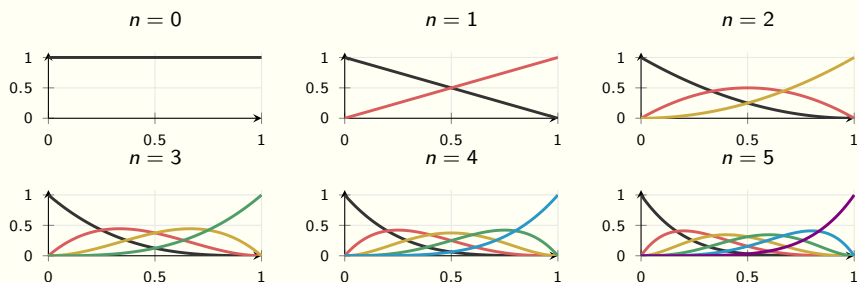
five Gaussian bumps a
third of a width apart
cond $\mathbf{G} = 2.4$ million

So far the shapes were orthogonal. Now take a family whose shapes overlap. Two unit shapes with overlap $\rho = \langle \phi_1, \phi_2 \rangle = 0.99$ point almost the same way, and their Gram matrix has condition number $(1 + \rho)/(1 - \rho) = 199$. Where does that number show up?

Their difference has length $\sqrt{2 - 2\rho} = 0.14$, so adding $0.1(\phi_1 - \phi_2)$ to a target moves the target by 0.014 and the coefficients by 0.1 and -0.1 .

Adding $0.1(\phi_1 + \phi_2)$ moves the target by 0.2 for the same coefficient change, fourteen times more, and $14.1 = \sqrt{199}$. Neighboring hats overlap with $\rho = 1/4$, and five Gaussian bumps a third of a width apart reach a condition number of 2.4 million.

Bernstein basis polynomials of degree 0 to 5



$$B_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}, \quad \sum_{i=0}^n B_{i,n}(t) = 1, \quad B_{i,n}(t) \geq 0$$

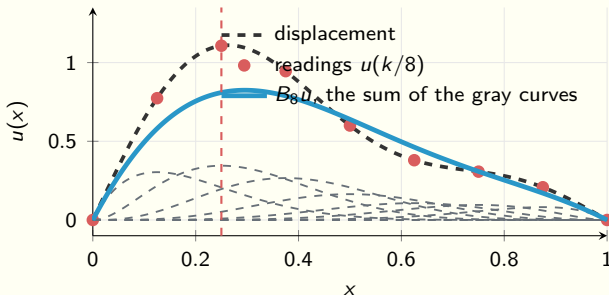
$B_{i,n}(t)$ is the probability of exactly i heads in n flips of a coin that lands heads with probability t

Before we use Bernstein's construction, look at the polynomials it averages with. The Bernstein basis polynomial of degree n and index i is $B_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}$, for t between 0 and 1 and $i = 0, \dots, n$, so there are $n+1$ of them. Each rises from zero to a single peak at $t = i/n$ and falls back to zero, except the first and last, which equal one at $t = 0$ and $t = 1$.

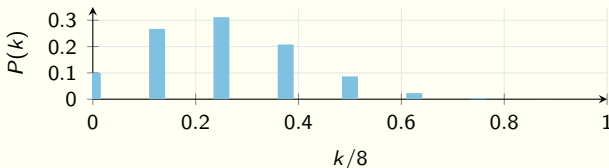
Read the formula as a coin. Take a coin that lands heads with probability t and flip it n times. The factor t^i is the probability of heads on i chosen flips, $(1-t)^{n-i}$ is the probability of tails on the rest, and $\binom{n}{i}$ counts the orders in which those heads and tails can occur, so $B_{i,n}(t)$ is the probability of exactly i heads.

The properties follow from the probabilities. The basis polynomials add to one at every t , because some number of heads must occur, and none is negative, because no probability is. Reversing the index reverses the coin, $B_{i,n}(t) = B_{n-i,n}(1-t)$, so the family is symmetric about $t = \frac{1}{2}$. A weighted sum $\sum_i \beta_i B_{i,n}(t)$ therefore averages the weights β_i , stays between the smallest and largest of them, starts at β_0 , and ends at β_n . With the weights as control points in the plane, this sum is a Bézier curve, the curve that vector drawing programs and scalable fonts use. Bernstein introduced these polynomials in 1912 to prove the Weierstrass theorem, and the next frames choose the weights so that the average approximates a function.

Bernstein polynomials for the string displacement



probability of k heads in 8 coins, $x = 0.25$



$$B_n u(x) = \sum_{k=0}^n u\left(\frac{k}{n}\right) b_{k,n}(x)$$

$$b_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}$$

$B_8 u(0.25)$, the weighted average

0.810 against
 $u(0.25) = 1.107$

largest gap, $n = 8, 32, 128$
0.30, 0.091, 0.024

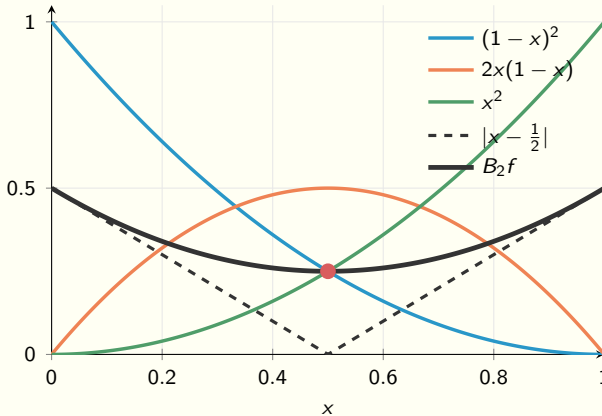
Voronovskaya, large n
 $3.18/n$

Can a family of smooth shapes come as close as we like to any continuous function, a corner included? Here is Bernstein's idea, tried first on a function it can handle, the string displacement. Take readings of the displacement at the nine points $k/8$.

To estimate the displacement at $x = 0.25$, flip eight coins that land heads with probability 0.25. The fraction of heads $k/8$ lands near 0.25, so average the readings with the probability of k heads as the weight. The weights are nonnegative and add to one, and the average is 0.810 against the true 1.107.

Do this at every x and the result is a polynomial of degree n , the sum of the gray curves $u(k/n)$ times the probability of k heads. Think of the reading $u(k/n)$ as the prize for k heads, so $B_n u(x)$ is the expected prize for a coin with bias x . With few flips the fraction of heads scatters widely, and the average mixes in readings from far away, which blurs the curve. With many flips the law of large numbers concentrates the fraction of heads around x , with spread $\sqrt{x(1-x)/n}$, and the average uses readings ever closer to x . How fast the gap falls depends on how much u changes over that spread. For a smooth displacement the poll is symmetric about x , so the first-order changes on either side cancel and the gap scales with the square of the spread, like $1/n$. The largest gap halves with each doubling of the coins, 0.30 at 8, 0.091 at 32, and 0.024 at 128. Voronovskaya's formula, $x(1-x)|u''(x)|/(2n)$, predicts $3.18/n$ for large n . No finite n reproduces the displacement, since Bernstein polynomials reproduce only straight lines exactly, and three sine weights reproduced it exactly. What happens at a corner?

Approximating a corner with Bernstein polynomials



the three hills at mid-span
 $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

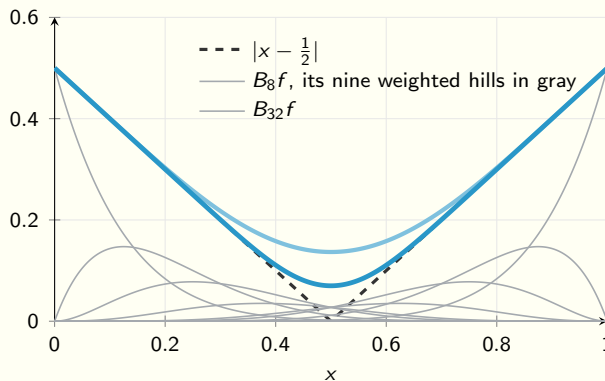
readings of $|x - \frac{1}{2}|$ at $0, \frac{1}{2}, 1$
 $\frac{1}{2}, 0, \frac{1}{2}$

B_2f at the corner
 $\frac{1}{4}$ where $f = 0$

Now the corner function $|x - \frac{1}{2}|$. The three quadratic Bernstein shapes are three hills, one at each wall and one at mid-span with height $\frac{1}{2}$, and they add to one at every x .

Weight each hill by a reading of the corner function $|x - \frac{1}{2}|$ at $0, \frac{1}{2}, 1$, that is $\frac{1}{2}, 0, \frac{1}{2}$, and add. At mid-span the hills weigh $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, so the sum reads $\frac{1}{4}$ where the function is 0. Every neighbor of the corner sits higher than the corner, so any average overshoots there, and the sum stays between the smallest and largest reading.

Convergence rate of Bernstein polynomials



$$B_n f(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$$

gap at the corner, $n = 2$
0.250

gap at the corner, $n = 8$
0.137, **predicted**
0.141

gap at the corner, $n = 32$
0.070, **predicted**
0.071

prediction
 $0.4/\sqrt{n}$

[Try it live](#)

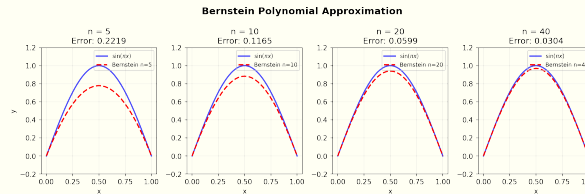
Here is Bernstein's idea. Flip n coins that land heads with probability x , and weight the reading $f(k/n)$ by the probability of exactly k heads. Each hill is that probability, peaked at k/n . With eight coins the hills narrow and the sum closes in. The gap at the corner is the average distance of the fraction of heads from one half, about $0.4/\sqrt{n}$, which predicts 0.141 against the measured 0.137.

With thirty-two the gap is 0.070 against a prediction of 0.071. The hills narrow like $1/\sqrt{n}$ and so does the gap, and it always sits at the corner, where the function is least smooth. At a corner the slope jumps, the first-order changes no longer cancel, and the gap scales with the spread itself, $0.4/\sqrt{n}$, so halving it takes four times the coins and one more correct digit takes a hundred times. The more a function varies over the spread of the poll, the more coins it needs, and the modulus of continuity measures that variation.

The Weierstrass approximation theorem

Theorem (Weierstrass, 1885)

For every continuous f on $[a, b]$ and every $\epsilon > 0$ there is a polynomial p with $|f(x) - p(x)| < \epsilon$ for every x in $[a, b]$.



Bernstein $B_n f$ for $\sin(\pi x)$,
largest gap
0.222, 0.117, 0.060,
0.030 at
 $n = 5, 10, 20, 40$

what the theorem does not
say
**the degree, the
weights, or the
slopes**

How many coin flips does a given accuracy need? Neither the theorem nor Bernstein's formula gives one number n in advance, and they differ in what they do give.

The Weierstrass theorem is an existence theorem. Give it a continuous function and a tolerance, say 0.01, and it guarantees that some polynomial comes within that tolerance everywhere on the interval. It names neither the polynomial, nor its coefficients, nor its degree.

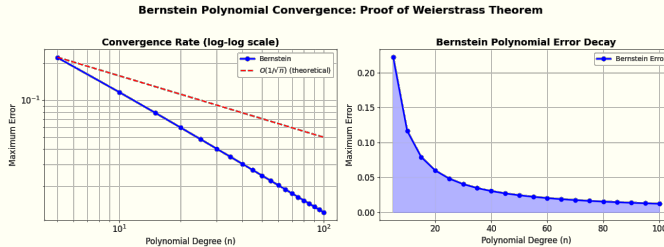
Bernstein's formula is a construction, $B_n f(x) = \sum_k f(k/n) \binom{n}{k} x^k (1-x)^{n-k}$, so its error can be computed for every n . The fraction of heads in n flips has variance $t(1-t)/n$, so the poll around t has width about $\sqrt{t(1-t)/n}$, and the error depends on how much the function changes over that width, which its modulus of continuity measures.

Take a gently sloping function first. Readings a little to either side of t are nearly equal, so a wide poll averages nearly equal readings, and a small n already lands close to $f(t)$. Now take a function that changes sharply over a short distance. A wide poll averages readings from both sides of the sharp change, and only a very narrow poll, a very large n , reads f near t alone.

For a merely continuous function the guaranteed rate is $1/\sqrt{n}$, and the corner meets it, so halving the error takes four times the flips and one more correct digit takes a hundred times. For a twice-differentiable function the symmetric poll cancels the first-order changes and the error falls like $1/n$, as the string and $\sin(\pi x)$ show, where each doubling of n halves the gap, 0.222, 0.117, 0.060, and 0.030 at $n = 5, 10, 20$, and 40. Either way Bernstein's polynomials converge slowly, and far better polynomials of the same degree exist. Every universal approximation theorem for networks has the form of Weierstrass's theorem, existence without the size.

Convergence of Bernstein polynomials for a smooth function

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



target
 $\sin(\pi x)$

largest gap, $n = 5$ and
 $n = 100$

0.22 and 0.013

Voronovskaya at mid-span,
 $\pi^2/(8n)$

0.25 and 0.012

For the corner the gap fell like $1/\sqrt{n}$. Give Bernstein a smooth target, $\sin(\pi x)$, and the error falls faster, along a slope of -1 on the log-log plot, from 0.22 at $n = 5$ to 0.013 at $n = 100$. Voronovskaya's formula predicts the gap for a twice-differentiable target, $f - B_n f \approx -x(1-x)f''(x)/(2n)$, which at mid-span has size $\pi^2/(8n)$, 0.25 at $n = 5$ and 0.012 at $n = 100$. So Bernstein approximation converges like $n^{-1/2}$ for this corner and like n^{-1} for this smooth sine. The dashed line in the figure is the general bound $O(1/\sqrt{n})$ for continuous targets, and the sine beats it. More smoothness does not give Bernstein more than n^{-1} , since even the analytic sine has a leading error of order $1/n$.

Least-squares polynomial approximation of $\sin(\pi x)$

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

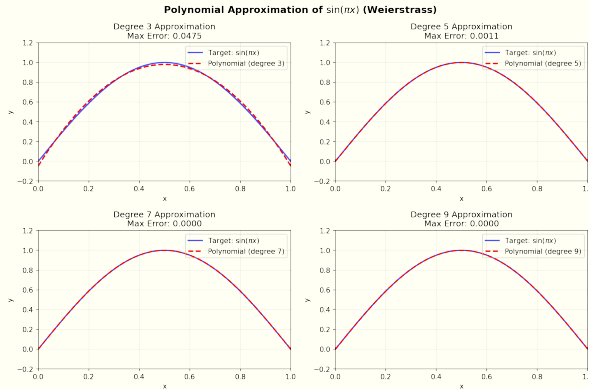
Noise

Families

Adaptive

The fin

Fields



degree 3, largest gap
0.0475

degree 5, largest gap
0.0011

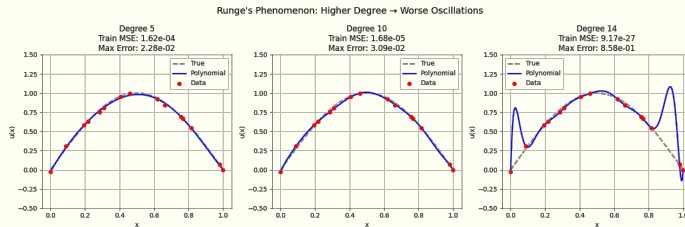
degrees 7 and 9, largest
gap
below 0.00005

Bernstein's polynomial is one polynomial of degree n , and a poor one. Fit $\sin(\pi x)$ by least squares on a hundred points instead. Degree 3 misses by 0.0475 in the sup norm, degree 5 by 0.0011, and degrees 7 and 9 by less than 0.00005.

For a function analytic on the interval, like the sine, the best polynomial error falls geometrically in the degree, while Bernstein's falls like $1/n$. The theorem promised only that some polynomial reaches any tolerance, and the least-squares fit reaches it at a far lower degree than Bernstein's construction.

High-degree interpolation of noisy data

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



degree 5, least squares
largest gap 0.023

degree 10, least squares
largest gap 0.031

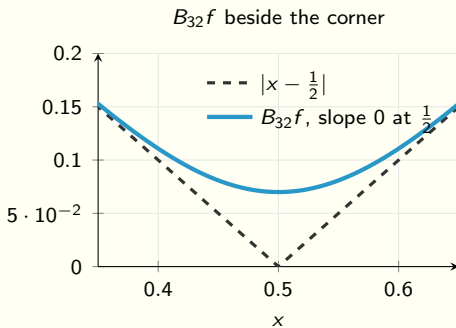
degree 14, through all
fifteen noisy readings
**training mean
squared error
 9×10^{-27} , largest
gap 0.86**

Now take fifteen scattered readings of the same sine, each with Gaussian noise of standard deviation 0.02, and fit them by least squares at degrees 5 and 10, and with the degree-14 polynomial through all fifteen points. The two least-squares fits miss by 0.023 and 0.031 in the sup norm. The interpolant passes through every noisy reading, with a root-mean-square residual near 10^{-13} , and misses the sine by 0.86 near the ends, where it swings between the data.

Between readings a high-degree polynomial is free to oscillate, and it amplifies the noise it was forced to reproduce, most near the ends of the interval. This experiment shows the instability of high-degree interpolation on noisy scattered data, which is related to Runge's divergence for noiseless equispaced interpolation and is a different experiment. The degree-seven polynomial through the eight noisy fin readings failed the same way, 31 percent against the truth.

What the Weierstrass theorem does not say

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



Bernstein's error
falls like $1/\sqrt{n}$

best degree- n polynomial
falls like $1/n$

the theorem names
**no polynomial, no
degree, no slope**

Polynomials come as close as we like to every continuous function. Notice what the theorem does not say. It names neither the polynomial nor its degree, and it says nothing about slopes, since every $B_n f$ has slope zero at the corner where the slope of f jumps from -1 to 1 . Bernstein's error falls like $1/\sqrt{n}$, the best degree- n polynomial does better, like $1/n$, and the theorem names neither.

Density is also different from having a closest member. Polynomials of a fixed degree contain their closest point, and the polynomials of all degrees have no closest member for the corner, since the distance can be made smaller than any positive number and is never zero.

Universal approximation theorems for neural networks have the same form, existence without a procedure that finds the approximation.

Proving density with detectors

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

Noise

Families

Adaptive

The fin

Fields

moments forced to zero		response to $\sin(3\pi x)$	$\Rightarrow \int_0^1 x^k d\mu(x) = 0$ for every k $\int_0^1 f d\mu = 0$ for every continuous f
	0	0.635	
	2	0.565	
	4	0.298	
	8	1.2×10^{-3}	
	16	3.0×10^{-9}	
	32	6.9×10^{-16}	

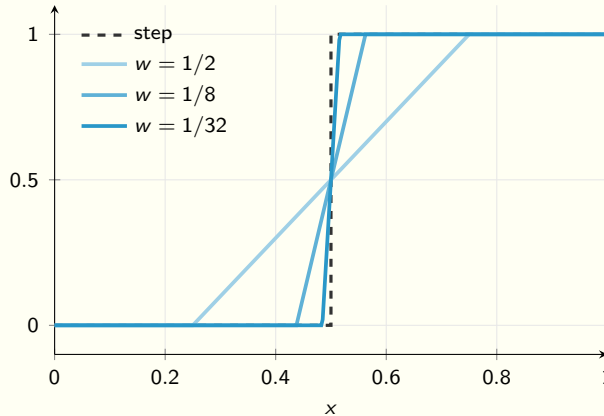
the density proof

a detector blind to every polynomial is blind to every continuous function

How is density proved? Run the claim backward. If polynomials could not reach some continuous f , a gap would separate f from the whole family, and functional analysis gives every gap a witness, a linear functional that vanishes on every polynomial yet responds to f . Vanishing on every polynomial means every moment is zero, $\int x^k d\mu = 0$ for all k , and proving density means showing such a detector responds to nothing.

The table is the finite-dimensional shadow of this argument, from the book's notebook. Take a sign-alternating weight vector on a grid as a candidate detector, force its first moments to zero by projection, and measure its response to $\sin(3\pi x)$. The response falls from 0.635 with no moments removed to 1.2×10^{-3} with eight and 6.9×10^{-16} with thirty-two. The universal approximation theorem for networks is proved with the same argument, with the polynomials replaced by the network's units.

Cauchy sequences, completeness, and Hilbert spaces



L^2 distance to the step,
 $\sqrt{w/12}$
0.204 at $w = 1/2$

$w = 1/8$
0.102

$w = 1/32$
0.051

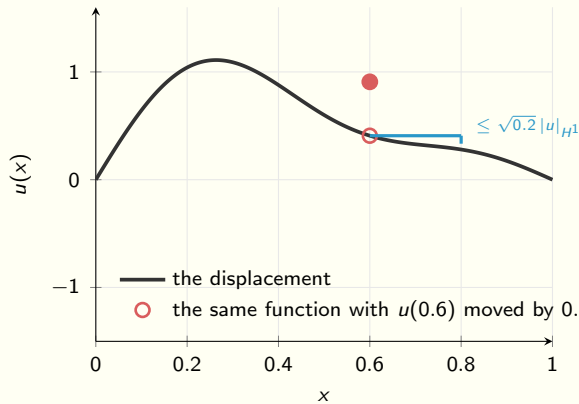
$\|f_8 - f_{32}\|_{L^2}$
0.077

A limit can lie outside a family in a second way. A continuous ramp of width w sits at L^2 distance $\sqrt{w/12}$ from the step.

Narrow the ramp. Every quartering of w halves the distance to the step.

The ramps also approach one another, $\|f_8 - f_{32}\| = 0.077$, a Cauchy sequence, and every ramp is continuous while the limit is the step. So the continuous functions with the L^2 distance are incomplete, and L^2 , where every Cauchy sequence converges, is their completion. A complete space with an inner product is a Hilbert space. The string's displacements form H_0^1 , the pinned functions of finite energy.

Point evaluation in L^2 and H^1

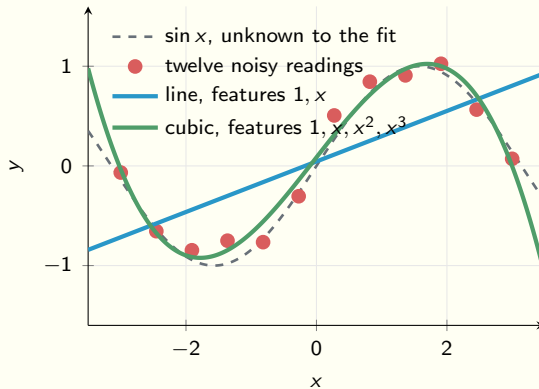


L^2 distance between the two
0

$$|u(y) - u(x)| = \left| \int_x^y u'(s) ds \right| \leq \sqrt{y-x} |u|_{H^1}$$

A sensor reads a value at a point, and L^2 cannot. The plot shows the displacement and a second function equal to it everywhere except at $x = 0.6$, where it reads 0.5 higher. The two have L^2 distance zero, so in L^2 they are the same element, and an element of L^2 has no value at a point. One derivative restores it. For a finite-energy displacement, $|u(y) - u(x)| = \left| \int_x^y u'(s) ds \right| \leq \sqrt{y-x} |u|_{H^1}$, so values at nearby points differ by at most the square root of the separation times the slope norm, and every element of H^1 on an interval has one continuous representative. Our station readings were legitimate because the string had finite energy. Now eight thermocouples on a cooling fin give eight such values, with noise, and nothing else. How do eight readings become a curve everywhere?

Linear regression by least squares



$$\hat{y}(x) = \sum_j w_j \phi_j(x), \quad \min_{\mathbf{w}} \|\Phi \mathbf{w} - \mathbf{y}\|$$

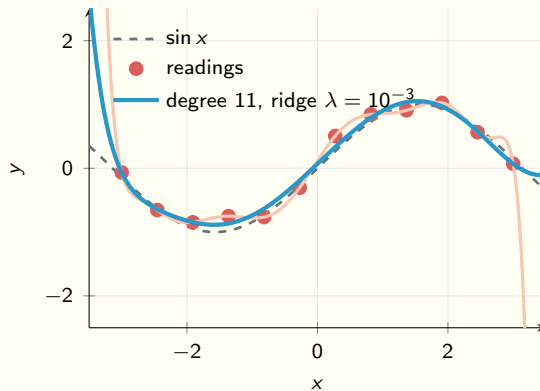
$$\Phi^\top \Phi \mathbf{w} = \Phi^\top \mathbf{y}$$

line, RMSE at the data
and against $\sin x$
0.484 and 0.439

cubic, the same two errors
0.119 and 0.085

Before kernels, the simplest fit to scattered readings. Here are twelve readings $y_i = \sin x_i + \varepsilon_i$ at evenly spaced points on $[-3, 3]$, with noise of standard deviation 0.15. Pick a few fixed feature functions ϕ_j , write the fit as their weighted sum, and choose the weights that minimize the squared misfit at the readings. With $\Phi_{ij} = \phi_j(x_i)$ the minimizer solves the normal equations $\Phi^\top \Phi \mathbf{w} = \Phi^\top \mathbf{y}$, which is the projection theorem again, the residual orthogonal to every feature. With the features 1 and x the fit is a line, and it misses by 0.484 at the readings and 0.439 against $\sin x$. With the cubic features $1, x, x^2, x^3$ the errors fall to 0.119 and 0.085. More features fit better, so why not use many?

Ridge regression on many features



$$\min_{\mathbf{w}} \|\Phi \mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|^2$$

$$(\Phi^T \Phi + \lambda \mathbf{I}) \mathbf{w} = \Phi^T \mathbf{y}$$

degree 11, least squares
error at the data 0,
largest weight 284,
swings to 29.5

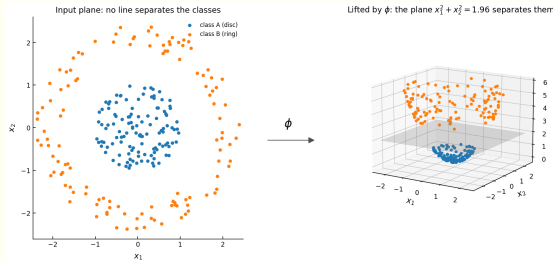
degree 11, ridge
error against sin x
0.073, largest weight
4.3

Give the fit twelve features, the powers of $x/3$ up to degree 11, as many as there are readings. Least squares then passes through every reading, noise included, with weights as large as 284, and the curve swings to 29.5 beyond the last readings. Fitting the readings exactly is the wrong goal when they carry noise.

Ridge regression adds a penalty on the size of the weights, $\lambda \|\mathbf{w}\|^2$, which changes the normal equations to $(\Phi^T \Phi + \lambda \mathbf{I}) \mathbf{w} = \Phi^T \mathbf{y}$. Adding λ to the diagonal lifts the small eigenvalues that amplified the noise. At $\lambda = 10^{-3}$ the largest weight falls to 4.3, the curve stays near the readings, and its error against $\sin x$ is 0.073, below the cubic's 0.085. The features were powers of x , chosen by hand. Which features suit a smooth profile, and do we need to write them down at all?

A kernel is an inner product of features

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



$$\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$$

$$k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x}) \cdot \phi(\mathbf{x}') = \mathbf{x} \cdot \mathbf{x}' + |\mathbf{x}|^2 |\mathbf{x}'|^2$$

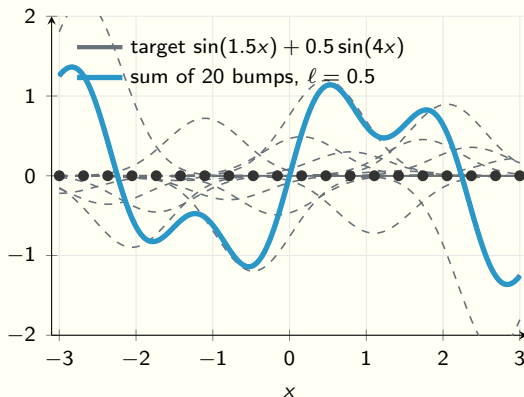
a plane in feature space
 $x_3 = 1.96$ separates
disc from ring, no
errors

$\mathbf{x} = (1, 0)$, $\mathbf{x}' = (0.5, 0.5)$
both routes give
1.000

Features can do more than bend a line. The points of a disc and a surrounding ring cannot be split by a line in the plane. Lift each point by $\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$, and the third coordinate is the squared radius, below 1 on the disc and above 3.25 on the ring, so the flat plane $x_3 = 1.96$ separates them with no errors.

A linear method in feature space only ever computes inner products $\phi(\mathbf{x}) \cdot \phi(\mathbf{x}')$. For this map the inner product has a closed form in the original coordinates, $\mathbf{x} \cdot \mathbf{x}' + |\mathbf{x}|^2 |\mathbf{x}'|^2$, and for $\mathbf{x} = (1, 0)$ and $\mathbf{x}' = (0.5, 0.5)$ both routes give 1.000. A kernel is such a function, $k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x}) \cdot \phi(\mathbf{x}')$, computed without forming ϕ . Its Gram matrix $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ is a matrix of inner products, so it is positive semidefinite, and the Gaussian kernel $e^{-(\mathbf{x}-\mathbf{x}')^2/2\ell^2}$ corresponds to an infinite list of features that it computes without ever writing them down.

Regression on Gaussian bump features



$$\hat{f}(x) = \sum_{j=1}^m w_j e^{-(x-c_j)^2/2\ell^2}$$

5 bumps, $\ell = 0.5$

RMSE 0.442

20 bumps, $\ell = 0.5$

RMSE 0.002

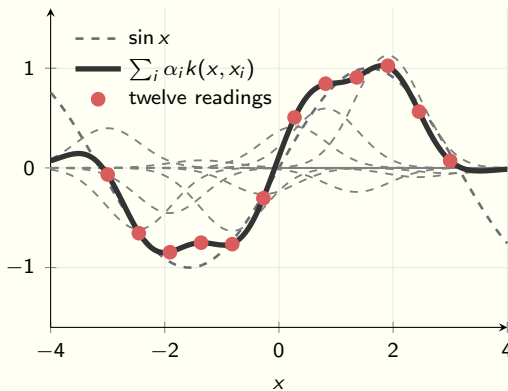
20 bumps, narrow $\ell = 0.1$
and wide $\ell = 2$

0.191 with gaps,
0.316 blurred

Now use Gaussian bumps as the features. Place m copies of one bump at evenly spaced centers c_j on $[-3, 3]$, give them the width ℓ , and fit their weights by least squares to the target $\sin(1.5x) + 0.5\sin(4x)$, with a tiny ridge term so the solve stays stable when the bumps overlap. This is linear regression again, with bump features.

Five bumps of width 0.5 sit 1.5 apart, too far to follow the fast wiggle, and the sum misses by 0.442 in root-mean-square. Twenty bumps of the same width reach 0.002. Twenty narrow bumps, $\ell = 0.1$, leave gaps between their centers and miss by 0.191, and twenty wide ones, $\ell = 2$, overlap so much that they blur the fast wiggle and miss by 0.316. The width has to match the spacing and the detail of the target, and the companion notebook, kernels/kernel-approximation, has the sliders.

Kernel ridge regression with copies at the data



$$\hat{f}(x) = \sum_{i=1}^{12} \alpha_i k(x, x_i), \quad (\mathbf{K} + \lambda \mathbf{I}) \alpha$$

weights, $\ell = 1/\sqrt{3}$,
 $\lambda = 0.01$

from -0.632 to 1.129

RMSE at the readings and
against $\sin x$

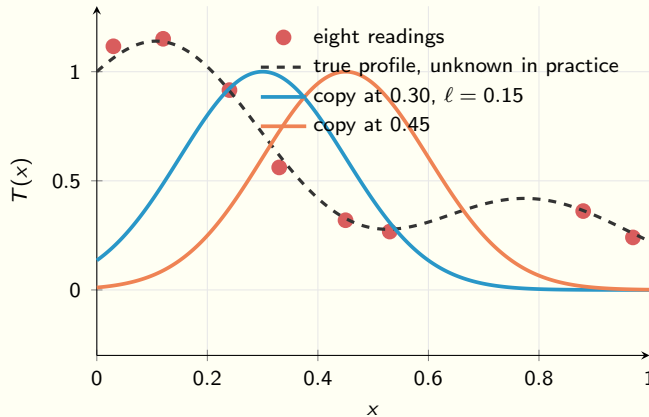
0.0051 and 0.113

Where should the bumps go? Put one copy at each reading, $\hat{f}(x) = \sum_i \alpha_i k(x, x_i)$, and fit the weights by ridge regression. The weights then solve $(\mathbf{K} + \lambda \mathbf{I}) \alpha = \mathbf{y}$ with $K_{ij} = k(x_i, x_j)$, the ridge equations with the kernel matrix in place of $\Phi^\top \Phi$. The representer theorem says this is no restriction, since the ridge minimizer over the whole space of the kernel's features is always a sum of copies at the data points.

Each gray dashed curve is one copy, centered at a reading and scaled by its weight, which ranges from -0.632 to 1.129 with $\ell = 1/\sqrt{3}$ and $\lambda = 0.01$. The black curve adds the twelve copies. It misses the readings by only 0.0051 and misses $\sin x$ by 0.113, close to the noise level 0.15, so at this λ the sum fits the noise. The rest of this section builds this machinery on the cooling fin, where the readings are all we have.

Kernels as similarity rules for scattered readings

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



$$k(x, x') = e^{-(x-x')^2/2\ell^2}$$

$$k(0.30, 0.45), \text{ one } \ell \text{ apart} \\ 0.61$$

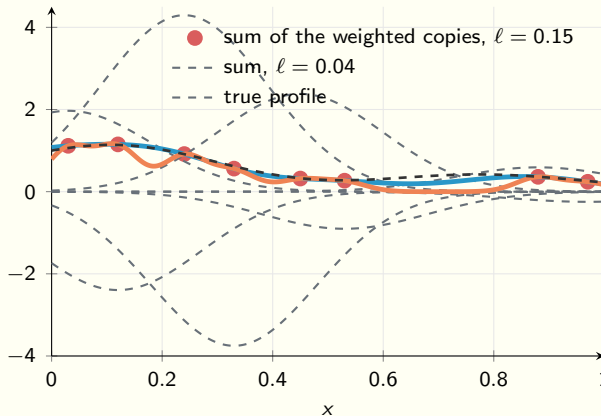
$$k \text{ at } 3\ell \text{ apart} \\ 0.011$$

For a cooling fin, only noisy temperature measurements are available, and the unknown curve must be inferred from those measurements and assumptions about its variation. Let $x \in [0, 1]$ denote normalized distance along the fin, and let y_i be the reading at x_i . The readings constrain the temperature at the sensors, but leave infinitely many possible curves between them. How should a reading influence the estimate at a neighboring location?

Heat conduction motivates a smooth temperature profile. Nearby locations should have similar temperatures, with less agreement expected as their separation grows. The squared-exponential kernel $k(x, x') = \exp(-(x - x')^2/2\ell^2)$ expresses this assumption, and ℓ sets the distance over which the similarity decreases. It is a parameter of the representation, whose suitability must be checked against the measurements. The copies centered at 0.30 and 0.45 are one length scale apart, with similarity $e^{-1/2} = 0.61$, and two locations 3ℓ apart have similarity 0.011.

A weighted sum of the copies gives the temperature estimate, $\hat{f}(x) = \sum_i \alpha_i k(x, x_i)$. Because neighboring copies overlap, the weights generally differ from the readings and must be chosen together.

Reconstructing a profile with weighted kernel copies



$$\mathbf{K}\alpha = \mathbf{y}, \quad K_{ij} = k(x_i, x_j)$$

weights at $\ell = 0.15$

alternate in sign, up to 4.30 and -3.75

two equal readings one ℓ apart

$1/(1 + \rho) = 0.62$ each

at $x = 0.7$, $\ell = 0.15$

0.216 against the reference 0.393

sum at $x = 0.7$, $\ell = 0.04$

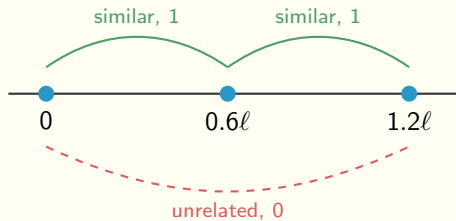
0.000 against the true 0.393

To interpolate the readings, evaluate the sum at each sensor and require $\hat{f}(x_i) = y_i$, which gives $\mathbf{K}\alpha = \mathbf{y}$ with $K_{ij} = k(x_i, x_j)$. For distinct sensor locations the squared-exponential kernel gives a symmetric positive-definite matrix, so the interpolation weights are unique. This guarantee concerns agreement at the sensors, and it says nothing about accuracy between them.

For a hand calculation, take two sensors separated by ℓ , each with reading one. Their similarity is $\rho = e^{-1/2}$, and symmetry gives equal weights, so at either sensor $\alpha + \rho\alpha = 1$ and each weight is $1/(1 + \rho) = 0.62$. The contribution from the neighboring copy reduces the weight needed at that sensor.

The fin readings here are synthetic, so a reference profile is available. At $\ell = 0.15$ the weights alternate in sign, up to 4.30 and -3.75 , and the interpolant at $x = 0.7$ is 0.216, against the reference value 0.393. At $\ell = 0.04$ the estimate there is below 0.001, although every sensor reading is still reproduced. Agreement at the sensors alone does not validate the length scale.

Positive definiteness of a kernel



$$\mathbf{w} = (1, -\sqrt{2}, 1), \quad \mathbf{w}^\top \mathbf{K} \mathbf{w} = -1.66$$

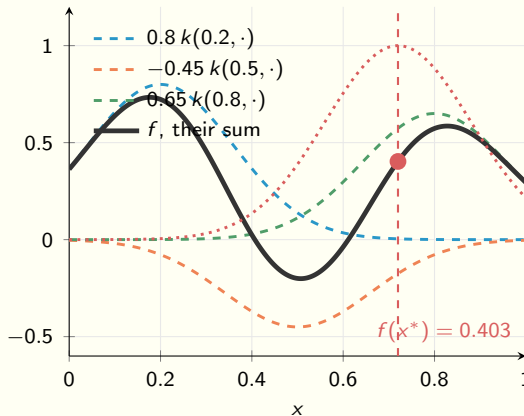
$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad \lambda = 1 - \sqrt{2}, 1, 1 + \sqrt{2}$$

$$k(x, x') = \phi(x)^\top \phi(x') \Rightarrow \mathbf{w}^\top \mathbf{K} \mathbf{w} = \left\| \sum_i w_i \phi(x_i) \right\|^2$$

The similarity between two locations should also describe the inner product of their kernel copies, and an arbitrary similarity rule need not support that. Consider a rule that assigns one to locations less than ℓ apart and zero otherwise. At the sites $0, 0.6\ell$, and 1.2ℓ it gives the matrix shown, and the weights $(1, -\sqrt{2}, 1)$ give $\mathbf{w}^\top \mathbf{K} \mathbf{w} = 4 - 4\sqrt{2} = -1.66$. If the entries were inner products, this value would be the squared norm of a weighted sum, and a squared norm cannot be negative, so this rule fails.

A symmetric kernel is positive definite when every finite choice of sites and weights gives a non-negative quadratic form. That convention permits zero, so the matrices are positive semidefinite, and strict positive definiteness requires a positive value for nonzero weights at distinct sites. The squared-exponential kernel satisfies the stronger condition. A kernel built from feature vectors, $k(x, x') = \langle \Phi(x), \Phi(x') \rangle$, gives $\mathbf{w}^\top \mathbf{K} \mathbf{w} = \left\| \sum_i w_i \Phi(x_i) \right\|^2 \geq 0$, and every positive-definite kernel admits such features, possibly infinite-dimensional and never written out. What function space does this inner product define?

The reproducing kernel Hilbert space



$$\langle k(x_1, \cdot), k(x_2, \cdot) \rangle_k = k(x_1, x_2)$$

$$\|f\|_k^2 = \alpha^\top \mathbf{K} \alpha, \ell = 0.16$$

1.041

$$f(x) = \langle f, k(x, \cdot) \rangle_k$$

$$f(0.72), \text{ read as an inner product}$$

0.403, **bound 1.02**

$$\text{slope bound}$$

$$|f'(x)| \leq \|f\|_k / \ell$$

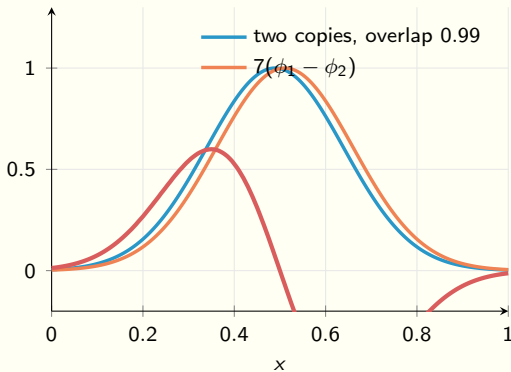
For the squared-exponential kernel a single copy has squared norm $k(x_i, x_i) = 1$, and copies at x_1 and x_2 have inner product $\rho = k(x_1, x_2)$. Their sum has squared norm $2 + 2\rho$ and their difference $2 - 2\rho$, so nearby copies are close in this norm. These calculations extend to every finite weighted sum by linearity, and completing the sums, so that their Cauchy sequences have limits, gives the reproducing kernel Hilbert space $\mathcal{H}_k$, with $\langle k(x_1, \cdot), k(x_2, \cdot) \rangle_k = k(x_1, x_2)$. This inner product differs from the integral of the product of two copies, 0.61 against 0.78 for unit bumps one length scale apart.

To evaluate a weighted sum, take its inner product with the copy centered at the evaluation point. For the member shown, three copies with $\ell = 0.16$ and squared norm $\alpha^\top \mathbf{K} \alpha = 1.041$, the copy at $x^* = 0.72$ returns $f(x^*) = 0.403$, and in general $f(x) = \langle f, k(x, \cdot) \rangle_k$. The Cauchy-Schwarz inequality bounds each point value, $|f(x)| \leq \|f\|_k \sqrt{k(x, x)}$, which is 1.02 here, so closeness in the RKHS norm implies closeness at every point, which the L^2 norm alone does not provide.

For this differentiable kernel the reproducing identity also gives the slope bound $|f'(x)| \leq \|f\|_k / \ell$. A function that increases by one over a distance $\ell/10$ must have a slope of at least $10/\ell$ somewhere, so its RKHS norm is at least ten. The norm constrains both amplitude and variation. How can this constraint help when the readings contain noise?

Noise amplification in kernel interpolation

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



eigenvalues of $\mathbf{K}$, overlap 0.99

1.99 along $(1, 1)$, 0.01 along $(1, -1)$

readings $(1, 1)$

weights near $\frac{1}{2}$ each

noise $(\varepsilon, -\varepsilon)$

weights $\pm\varepsilon/0.01$, a hundred times the noise

$\|\phi_1 - \phi_2\|$

0.14, so the curve moves fourteen times the noise

Interpolation reproduces the noise along with the measured temperature, and the effect can be large when nearby kernel copies are nearly dependent. For two sensors with similarity $\rho = 0.99$ the matrix eigenvalues are $1 + \rho = 1.99$ and $1 - \rho = 0.01$, along equal and opposite changes in the readings.

Opposite measurement errors $(\varepsilon, -\varepsilon)$ change the interpolation weights by $(100\varepsilon, -100\varepsilon)$. The difference of the two copies has RKHS norm $\sqrt{2 - 2\rho}$, so the reconstructed function changes by about $14.1|\varepsilon|$ in RKHS norm. Large coefficient changes and large function changes are related, and their magnitudes differ.

Kernel ridge regression and the choice of λ

$$\min_{f \in \mathcal{H}_k} \sum_{i=1}^n (f(x_i) - y_i)^2 + \lambda \|f\|_k^2$$

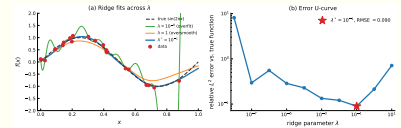
$$(\mathbf{K} + \lambda \mathbf{I})\alpha = \mathbf{y}$$

$\lambda = 0.1$, small divisor

0.01 $\rightarrow$ 0.11

amplification

100 to 9.1



A regularized fit permits disagreement with noisy readings and penalizes the RKHS norm. For $\lambda > 0$, kernel ridge regression minimizes $\sum_i (f(x_i) - y_i)^2 + \lambda \|f\|_k^2$ over $\mathcal{H}_k$. The first term measures squared disagreement at the sensors, the second discourages the amplitude and variation needed to fit every fluctuation, and λ sets their relative importance.

Any component orthogonal to the span of the sensors' kernel copies has zero value at every sensor, so it changes none of the fitting errors and increases the norm, and the minimizer has no such component. The fit therefore remains a weighted kernel sum, with $(\mathbf{K} + \lambda \mathbf{I})\alpha = \mathbf{y}$. Adding λ to each eigenvalue reduces the amplification caused by small eigenvalues, and in the two-sensor example $\lambda = 0.1$ changes the small divisor from 0.01 to 0.11, so the amplification falls from 100 to about 9.1.

The figure sweeps twenty noisy sine samples across nine decades of λ . Near-interpolation reaches ninety times the best error, the minimum sits near 10^{-1} , and over-smoothing rises beyond it, so in practice λ is chosen by predicting held-out readings. Can the same kernel also describe uncertainty between the sensors?

Conditioning a Gaussian on an observation

Sampling

Vectors

Norms

Spaces

Projection

Bases

Convergence

Kernels

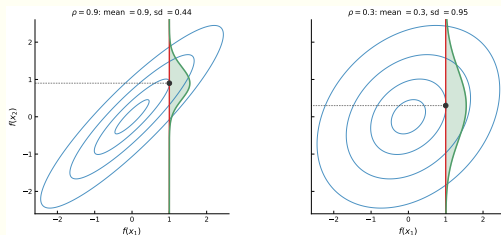
Noise

Families

Adaptive

The fin

Fields



$$f(x_1) = Z_1, \quad f(x_2) = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$$

$$f(x_2) \mid f(x_1) = y \sim \mathcal{N}(\rho y, 1 - \rho^2)$$

$$\rho = 0.9$$

**mean 0.9y, standard
deviation 0.44**

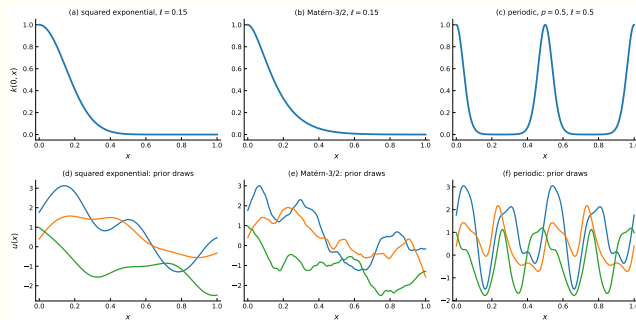
$$\rho = 0.3$$

**mean 0.3y, standard
deviation 0.95**

A temperature at an unmeasured location can be described by a probability distribution. To see how a measurement changes that distribution, take independent standard Gaussian variables Z_1 and Z_2 and define $f(x_1) = Z_1$ and $f(x_2) = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$. Both temperatures have mean zero and variance one, with covariance ρ .

Observing $f(x_1) = y$ fixes $Z_1 = y$, and the remaining temperature has conditional mean ρy and standard deviation $\sqrt{1 - \rho^2}$, which is 0.44 at $\rho = 0.9$ and 0.95 at $\rho = 0.3$. Stronger correlation gives less remaining uncertainty. A Gaussian process extends this construction to any finite collection of locations, with zero prior mean and covariance $k(x, x')$.

Prior draws from three kernels



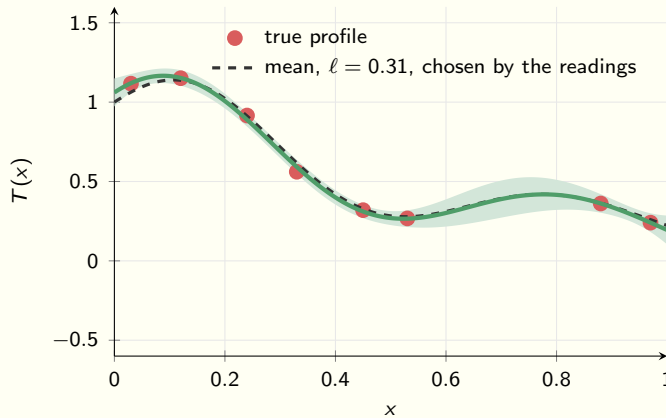
mean absolute slope of the
draws
5.2, 8.9, and 20

a kernel method returns
**functions that
resemble its draws,
pulled toward the
data**

A Gaussian process makes any finite list of values jointly Gaussian, and sampling its prior shows which profiles it permits. Squared-exponential draws are infinitely smooth, Matérn-3/2 draws have continuous values and rough slopes, and periodic draws repeat exactly. At the plotted parameters the mean absolute slopes over five hundred draws are 5.2, 8.9, and 20.

The kernel is the prior. Its draws are its beliefs, and the posterior is those beliefs conditioned on the readings. Typical draws differ from the fitted posterior mean, which lies in the span of the n kernel copies.

Gaussian-process reconstruction of the fin profile



mean
 $\mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$

variance
 $k(x, x) - \mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(x)$

$\ell = 0.31$
error 3.2 percent,
half-width in the gap
0.12

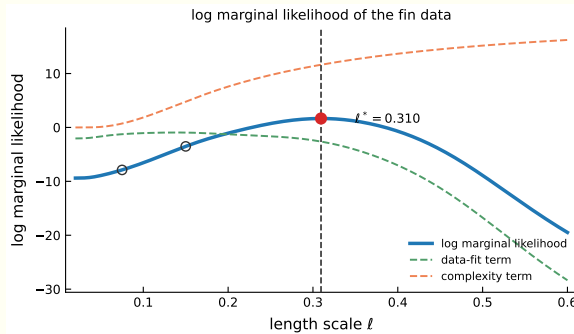
value at $x = 0.7$
0.389 against the
true 0.393

The measured values are $y_i = f(x_i) + \varepsilon_i$, with independent Gaussian noise of variance σ^2 , and conditioning gives the mean $m(x) = \mathbf{k}_x^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$ and the variance $v(x) = k(x, x) - \mathbf{k}_x^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_x$. The mean equals the kernel ridge fit with $\lambda = \sigma^2$ for the unnormalized squared-error objective. The variance measures remaining uncertainty about the latent temperature and excludes the noise of a future measurement, and the subtracted term is the reduction supplied by the sensors.

The shaded band is $m(x) \pm 2\sqrt{v(x)}$, conditional on the selected kernel and noise model. With noise standard deviation 0.03, its largest half-width in the unsensed stretch is 0.90 at $\ell = 0.15$ and 1.99 at $\ell = 0.075$, so shorter-range correlations leave greater uncertainty between the sensors. These are pointwise bands, without a guarantee of simultaneous coverage of the whole curve.

Maximizing the marginal likelihood over the tested length scales gives $\ell = 0.31$ in this synthetic experiment, where the mean has relative L^2 error 3.2 percent and the largest band half-width in the gap is 0.12. These results depend on the model assumptions, which must be reconsidered if the temperature has a sharp transition.

Choosing the length scale by marginal likelihood



log marginal likelihood at
 $\ell = 0.31$

1.7, the maximum

at $\ell = 0.15$ and 0.075

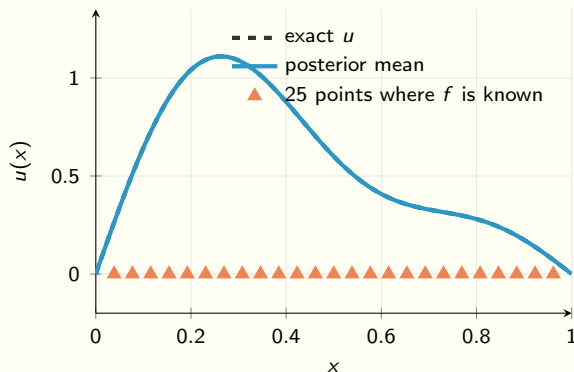
-3.5 and -7.9

with eight readings

a soft verdict

The readings also provide evidence about the kernel's length scale and the noise level. The marginal likelihood is the probability the prior-plus-noise model assigns to the readings seen, and it trades a data-fit term against a complexity term. A short length scale fits the readings easily and is penalized by the complexity term, and a long one is simple and misses the readings. For the fin the maximum sits at $\ell = 0.31$, with log marginal likelihood 1.7, against -3.5 at 0.15 and -7.9 at 0.075. With only eight readings the curve is broad, so the selected length scale is itself uncertain, and the band's dependence on ℓ is part of what the model does not know.

The kernel prior solves $\mathcal{L}u = f$



$$\text{cov}(u, f) = \mathcal{L}_{x'} k$$
$$\text{cov}(f, f) = \mathcal{L}_x \mathcal{L}_{x'} k$$

relative L^2 error of the mean

$$3.2 \times 10^{-7}$$

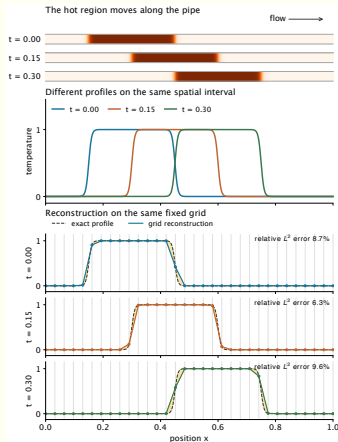
condition number of the observation covariance

$$1.4 \times 10^{12}$$

Now return to the string and its equation. Linear operators pass through Gaussian priors. Place a squared-exponential prior with $\ell = 0.2$ on the displacement u , apply $\mathcal{L} = -d^2/dx^2$ through the kernel to get the covariance of u with f and of f with itself, and condition on the known load at twenty-five points and the two pinned ends.

No training occurs, and the posterior mean solves $-u'' = f$ for the test displacement to a relative error of 3.2×10^{-7} . The method works only for linear operators, and its limit is numerical, the condition number of the observation covariance, 1.4×10^{12} . The same conditioning that reconstructed the fin from readings solves the equation from the load.

A moving front gives a family of functions



$$T(x, t) = T_0(x - ct), \quad \mathcal{S} = \{T_0(\cdot - ct) \mid 0 \leq t \leq 1\}$$

$$\hat{T}(x, t) = \sum_i T(x_i, t) \phi_i(x)$$

32 fixed nodes, times 0,
0.15, 0.30
8.7, 6.3, 9.6 percent

edge width against node
spacing
0.01 against 1/31

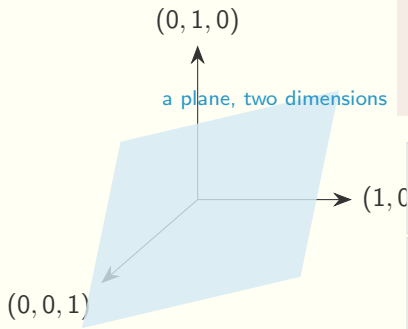
Now consider a hot region transported along a pipe. For constant flow speed c and negligible diffusion, its temperature profile translates without changing shape, and at each time the profile is a function of position on the same interval. The reconstruction must describe the hot region wherever it moves.

The figure shows the hot region in the pipe, its temperature profiles, and their reconstructions on one fixed grid, where the edges move to new locations while the grid nodes remain in place. The profile has transition width $w = 0.01$ and moves at normalized speed $c = 1$, and the grid has 32 nodes with spacing $1/31$. At times 0, 0.15, and 0.30 its relative L^2 reconstruction errors are 8.7, 6.3, and 9.6 percent, so the same grid represents some edge locations more accurately than others.

Translation gives $T(x, t) = T_0(x - ct)$, and each time selects a different function of x , while T_0 and c remain fixed. The collection $\mathcal{S}$ is a family parameterized by time, and knowing the initial shape and the translation rule specifies every member from that parameter. A fixed-grid representation changes only the coefficients of its hats with time, so every reconstructed profile belongs to the same linear span, and many nodes are needed when the edge width is small against their spacing. Does choosing a different fixed basis remove this difficulty for the whole family?

The Kolmogorov n -width

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



Kolmogorov n -width

$$d_n(\mathcal{M}) = \inf_{\dim V=n} \sup_{f \in \mathcal{M}} \inf_{v \in V} \|f - v\|$$

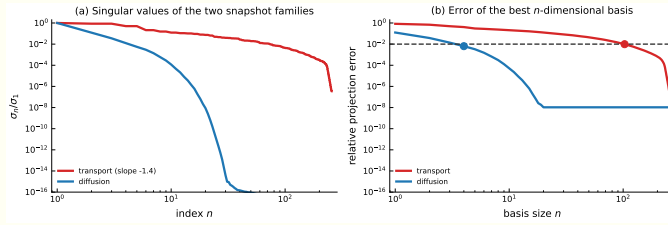
three unit spikes, squared
distances to any plane
sum to $3 - 2 = 1$

worst spike
at least
 $\sqrt{1/3} = 0.577$

A narrow pulse centered at separated locations gives profiles with little overlap, and the finite-dimensional version is a unit spike translated across three grid points. Its snapshots are the perpendicular vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in $\mathbb{R}^3$. Let P be the orthogonal projection onto any plane through the origin. The squared projection lengths add to the plane's dimension, so the squared errors add to $3 - 2 = 1$, and at least one error is no smaller than $1/\sqrt{3} = 0.577$. No plane approximates all the snapshots more closely, and exact linear representation needs dimension three, although one position parameter selects the spike.

For a family of functions, the Kolmogorov n -width asks how well the best fixed n -dimensional linear space approximates its worst member. The inner infimum chooses the closest member of V for each target, the supremum chooses the hardest target in the family, and the outer infimum chooses the space, which must be chosen once for the entire family.

Widths of diffusion and transport families



diffusing pulse, SVD, one percent mean-square snapshot error

four modes

translating pulse, same measure

103 modes

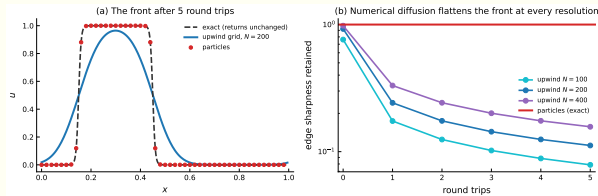
coefficient k under diffusion
shrinks by $e^{-(2\pi k)^2 t}$

coefficient k under translation
changes phase only

Motion alone does not imply a large width. Every translate of $\sin(kx)$ lies in the span of $\sin(kx)$ and $\cos(kx)$ by the angle-subtraction identity. A localized pulse with steep edges instead requires many frequencies, and translation does not reduce their magnitudes, so the translating pulse needs 103 modes for one percent. For a periodic heat equation with unit diffusivity those magnitudes decay by $e^{-(2\pi k)^2 t}$, which is why the diffusing pulse needs only four.

A snapshot matrix's singular value decomposition minimizes the average squared projection error over the sampled profiles, while the width concerns the worst profile over the entire family. A small average error on snapshots therefore needs a separate check of poorly represented positions. Can the representation adapt to the target instead of using one fixed linear space?

Particles carry values with the flow



markers after five round trips

the front returns unchanged

upwind grid at 100, 200, 400 points

keeps 8, 11, and 16 percent of the edge steepness

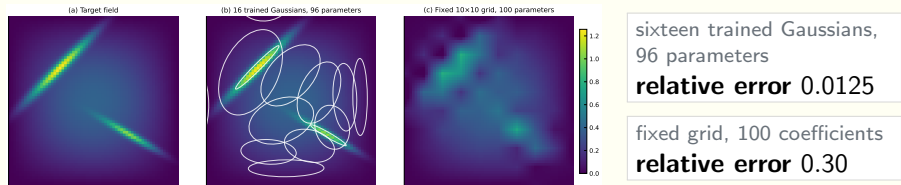
The translated-template formula already gives an alternative to fixed-grid coefficients. It stores the initial shape and evaluates that shape at shifted coordinates, which assumes a known translation law, and its functions need not belong to one low-dimensional linear space. Other adaptive representations change their locations, shapes, or selected terms.

A particle representation stores positions together with temperature values. For pure advection each particle keeps its temperature while its position follows the flow, so the front returns after five round trips unchanged. The field between particles is still reconstructed from their values by an interpolation rule, and exact transport of stored values does not remove that reconstruction error.

A first-order upwind grid scheme gives the comparison. For positive speed and $0 < \nu = c\Delta t/h < 1$ its update is $T_i^{n+1} = (1 - \nu)T_i^n + \nu T_{i-1}^n$, and each step averages neighboring temperatures, which spreads a steep transition. It keeps eight, eleven, and sixteen percent of the edge steepness at 100, 200, and 400 points. This numerical diffusion comes from the update rule, separately from the fixed basis's approximation error, and at $\nu = 1$ the update shifts grid values without averaging.

Gaussian splats with trainable centers and shapes

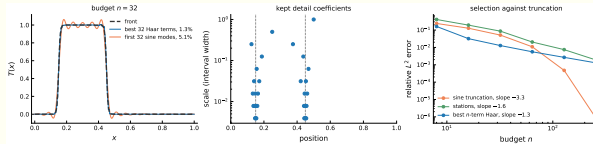
Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



A Gaussian element describes a localized field through its center, covariance, and amplitude, and an anisotropic covariance permits a long, narrow shape aligned with a ridge. Fitting these parameters moves the elements to where the field varies. Sixteen trained Gaussians, ninety-six parameters, reach a relative L^2 error of 0.0125 on a field with two thin ridges, against 0.30 for a fixed grid of one hundred coefficients.

With centers and covariances fixed, a quadratic fitting objective gives a linear system for the amplitudes. Fitting the centers and covariances as well generally makes the objective non-convex, so the representation can adapt to localized features while the fitting algorithm may stop at a poor local minimum.

Selecting coefficients from a fixed dictionary



$n = 16$, sines, stations,
largest Haar
12.8, 19.3, 3.2
percent

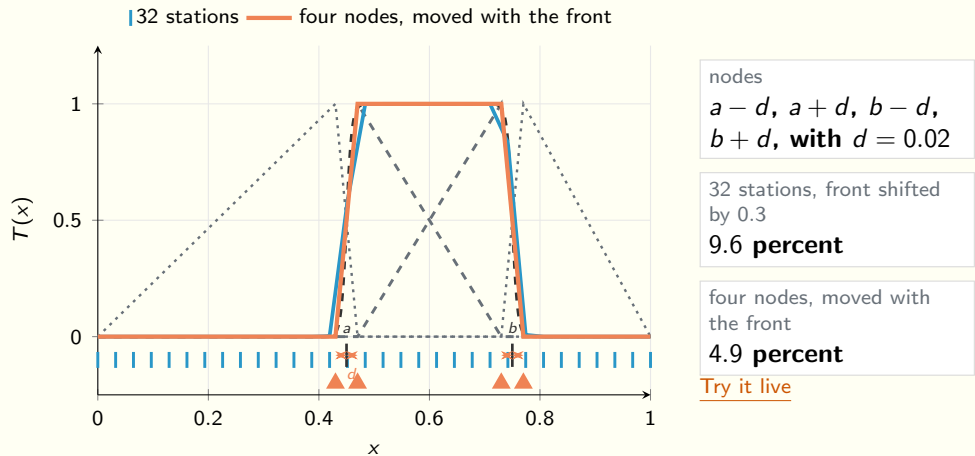
$n = 32$, largest Haar
against sines
1.3 against 5.1
percent

sines overtake
near $n = 128$

A wavelet representation keeps its dictionary fixed and selects terms for each profile. For an orthonormal Haar expansion, retaining the n largest coefficients minimizes the L^2 truncation error among selections of n terms. For a front the selected terms concentrate near the edges, and as the edges move, different terms are selected.

At $n = 16$ the first sixteen sine modes leave 12.8 percent, sixteen stations leave 19.3, and the sixteen largest Haar coefficients leave 3.2, and at $n = 32$ the Haar selection reaches 1.3 against 5.1 for the sines. Near $n = 128$ the sine truncation resolves the edge width and the curves change order. The selected functions belong to different dictionary subspaces, so the width's fixed-space restriction no longer describes this approximation. Storage must include the selected indices as well as the coefficient values, and resolution and truncation still limit accuracy.

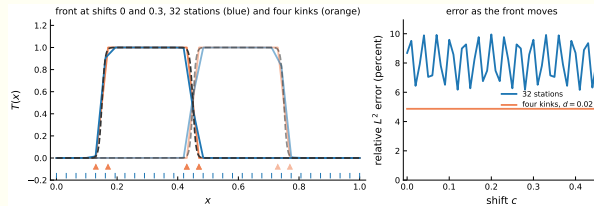
Movable hat nodes on a traveling front



The nodes of a hat basis need not sit on an even grid. Where would we want them? Put four nodes on the front, at $a - d$, $a + d$, $b - d$, and $b + d$, where a and b are the edges and $d = 0.02$. Each node carries a hat that peaks there and reaches zero at its neighbors. The hats at $a + d$ and $b - d$ carry weight one (dashed), and the hats at $a - d$ and $b + d$ carry weight zero (dotted). Their sum rises across the first edge, stays at one, and falls across the second, and it rebuilds the front at 4.9 percent, against 8.7 for thirty-two stations. A larger d widens the ramps, and at $d = 0.06$ the error is 19.8 percent.

Now move the front. The stations stay where we put them and their error rises to 9.6 percent, while the four nodes follow the edges and keep 4.9. The nodes use four numbers where the stations use thirty-two.

Fixed nodes and moving nodes



fixed nodes

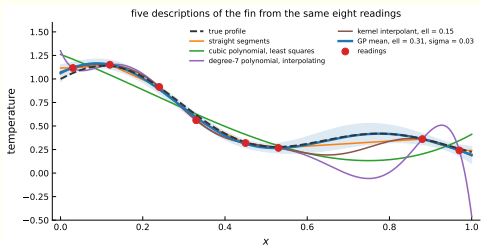
**a linear solve, with
the projection
theorem's guarantee**

moving nodes

**a non-convex fit,
with no guarantee**

Fixed nodes make the family a linear subspace, so fitting it is one solve, and the projection theorem guarantees that the solve finds the unique closest member. Moving nodes give a non-convex problem, and no such guarantee, the choice the splats already made. The gain is on fields whose features sit in a few places, the loss is the guarantee, and the width of a family says how much a fixed basis had to lose.

Choosing a representation for the cooling fin



description	error vs truth	l
straight segments	6.5 percent	
cubic, least squares	23	
degree-7 interpolation	31	
kernel interpolant, $\ell = 0.15$	11	
Gaussian-process mean, $\ell = 0.31$	3.2	
noise level		

For the fin, the unknown profile must be reconstructed from the same noisy measurements. The synthetic reference permits a direct comparison, but such a reference is unavailable in practice, so the table compares the error against that reference with the prediction error obtained by withholding readings.

The held-out calculation removes each reading, fits the remaining readings, and predicts the omitted value, with its root-mean-square error measured against the noise standard deviation 0.03. The length scales remain fixed at values selected using all the readings, so this is a fixed-hyperparameter diagnostic, and testing the complete selection procedure requires choosing the length scale again within each fold.

The table favors the Gaussian-process fit in this experiment, without establishing that it is best for every fin. The kernel specifies the possible functions and their geometry, regularized fitting selects a function using the measurements, and the posterior band adds uncertainty conditional on the same assumptions.

Stored numbers, shape placement, and assumption for each family

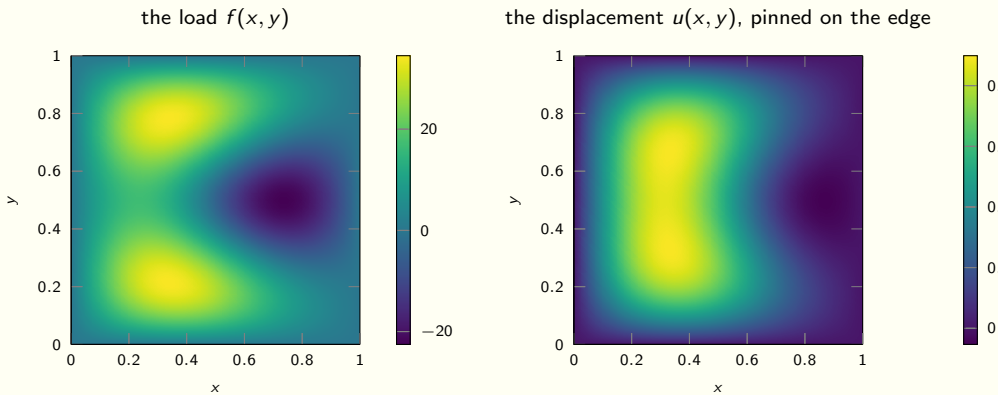
Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields

family	stores	shapes placed by	assumes
stations, grid	values at fixed points	analyst	linear variation between stations
mesh hats	nodal values	analyst and mesher	piecewise-linear variation
sine modes	global coefficients	analyst	smooth odd periodic continuation
B-splines, NURBS	coefficients at fixed knots	analyst and CAD	smoothness across knots
wavelets	selected coefficients and indices	analyst, data selects	few active scales per location
POD modes	coefficients on data-derived modes	the snapshot data	a family near a low-dimensional subsp
sparse dictionary	few atoms and their indices	the optimizer	few active atoms
particles, SPH, MPM	positions and payloads	the dynamics	payloads ride their carriers
Gaussian splats	centers, covariances, amplitudes	the optimizer	smooth field with concentrated featur
level sets	values of an implicit function	analyst and physics	interface as a zero set
kernels, GP	weights, sites, kernel parameters	analyst and data	regularity set by the kernel
neural fields	network weights	the optimizer	the network's smoothness bias
latent decoders	codes and a decoder	the optimizer	a low-dimensional nonlinear manifold

The fin comparison separates the stored numbers, the reconstruction rule, and the fitting principle, and the same distinctions apply to every representation in this table, including ones developed in later chapters.

Storage includes whatever the reconstruction rule cannot supply in advance. A uniform grid specifies its locations by a rule, while particles store their current positions, selected wavelets store indices, and POD coordinates need their stored modes. The fitting principle supplies guarantees under stated conditions, and those conditions are the first thing to check when a fit goes wrong. For stations it is aliasing, for sines the overshoot beside a jump, for kernels a length scale wrong for the field, and for splats a descent that stalls. The string supplied known functions for studying approximation, while the fin supplied observations for studying reconstruction.

A membrane under a load, $-\Delta u = f$

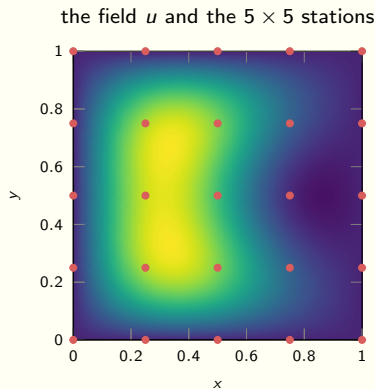


$$-\Delta u = -\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = f \text{ on the unit square,} \quad u = 0 \text{ on its edge}$$
$$u = 0.8 \sin(\pi x) \sin(\pi y) + 0.4 \sin(2\pi x) \sin(\pi y) + 0.2 \sin(\pi x) \sin(3\pi y)$$

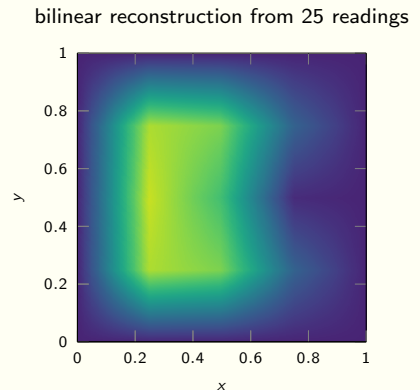
So far every function lived on the interval $[0, 1]$. A field is a function on a region of space, such as the temperature $T(x, y)$ on a plate or the displacement $\mathbf{u}(x, y, z)$ in a solid, a vector with three components. The string's two-dimensional version is a membrane stretched over the unit square and pinned on its edge, where a load $f(x, y)$ produces a displacement that satisfies $-\Delta u = f$. Take a test displacement built from three modes of the membrane, $0.8 \sin(\pi x) \sin(\pi y) + 0.4 \sin(2\pi x) \sin(\pi y) + 0.2 \sin(\pi x) \sin(3\pi y)$, which peaks at 0.900. The load that produces it is $f = -\Delta u$, which peaks at 34.61, since each mode $\sin(j\pi x) \sin(k\pi y)$ is multiplied by $(j^2 + k^2)\pi^2$. The field needs the same two things as a function, a list of numbers and a rule that rebuilds it, and every representation of the lecture carries over, with the coordinate x replaced by a point of the region.

Readings on a grid and bilinear reconstruction

Sampling
Vectors
Norms
Spaces
Projection
Bases
Convergence
Kernels
Noise
Families
Adaptive
The fin
Fields



5×5 , 25 values
largest gap 22.2 percent



9×9 , 81 values
7.1 percent

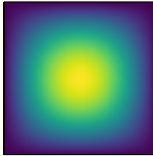
17×17 , 289 values
1.9 percent

The first representation is the same as for the string. Put stations on a grid, 5 in each direction, and record the displacement at the 25 nodes. Inside each square cell, bilinear interpolation weights the four corner readings so that the surface is straight along every line parallel to an edge.

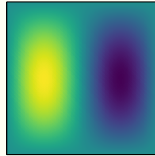
The reconstruction from 25 readings misses by 22.2 percent of the largest displacement. With 9 per direction, 81 values, it misses by 7.1 percent, and with 17, 289 values, by 1.9 percent, so each halving of the spacing still divides the gap by about four. The count is what changed. Seventeen stations per direction are 17 numbers on a line, 289 on a plate, and 4,913 in a solid, so for a given spacing the count on a line is raised to the power of the dimension.

Sine modes and hats in two dimensions

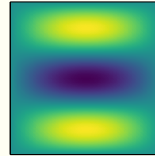
$\sin(\pi x) \sin(\pi y)$



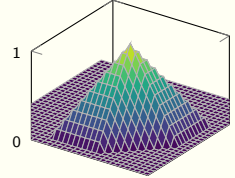
$\sin(2\pi x) \sin(\pi y)$



$\sin(\pi x) \sin(3\pi y)$



a hat on a triangle mesh



$$u(x, y) = \sum_{j,k} a_{jk} \sin(j\pi x) \sin(k\pi y), \quad u_h(x, y) = \sum_i u(\mathbf{x}_i) \phi_i(x, y)$$

the membrane's weights
 $a_{11} = 0.8, a_{21} = 0.4,$
 $a_{13} = 0.2$

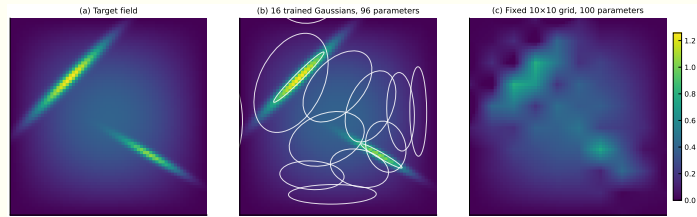
a hat ϕ_i

**one at its node, zero
at the others, linear
on each triangle**

Sine modes become products, $\sin(j\pi x) \sin(k\pi y)$, one coefficient for each pair (j, k) , and the membrane is three of them with weights 0.8, 0.4, and 0.2. They are orthogonal on the square, so each weight is still one inner product, and they vanish on the whole edge, so every sum is pinned.

Hats become pyramids on a triangle mesh. A hat equals one at its node and zero at every other node, and it is linear on each of the triangles around its node, so the sum of hats weighted by the readings is the surface through them made of flat triangular pieces. Triangles can follow a curved or irregular boundary, which a square grid cannot, and that is why finite elements use them. Galerkin's method with these hats solves $-\Delta u = f$ in the same way it solved the string, from the weak form $\int \nabla u_h \cdot \nabla \phi_i = \int f \phi_i$.

What changes in two and three dimensions



17 stations per direction
17 on a line, 289 on a plate, 4,913 in a solid

a kernel needs only a distance
 $e^{-|\mathbf{x}-\mathbf{x}'|^2/2\ell^2}$,
unchanged

H^1 in two or more dimensions
finite energy, yet possibly unbounded

vector fields
constraints such as
 $\nabla \cdot \mathbf{u} = 0$

Some representations carry over with no change. A kernel needs only a distance, so $k(\mathbf{x}, \mathbf{x}') = e^{-|\mathbf{x}-\mathbf{x}'|^2/2\ell^2}$ works unchanged, and scattered sensors on a plate are no harder than on a fin. Splats, particles, and neural fields already live in several dimensions, and the two-ridge field on the left is a two-dimensional field fitted with sixteen trained Gaussians.

Other things get harder. The number of values for a fixed spacing grows like a power of the dimension, 17 on a line, 289 on a plate, and 4,913 in a solid, which is the main reason adaptive and learned representations matter for fields. The function spaces change as well. In one dimension a displacement of finite energy is continuous, and in two or more dimensions an H^1 field can be unbounded at a point, so a point reading is no longer guaranteed to make sense and sensors measure averages over a small region. A vector field can also carry a constraint, such as zero divergence for an incompressible flow, that the representation has to respect.

Summary

- 1 Pointwise sampling
- 2 Functions as vectors
- 3 Norms and the meaning of error
- 4 Function spaces, L^2 , H^1 , and Hilbert spaces
- 5 Inner products, projection, and orthogonality
- 6 Changing the basis
- 7 Convergence, density, and completeness
- 8 Kernels and point evaluation
- 9 Noisy observations, ridge regression, and Gaussian conditioning
- 10 Approximating a family of functions
- 11 Adaptive representations
- 12 Choosing a representation for the cooling fin
- 13 From functions to fields

A representation stores numbers and specifies how to reconstruct a function, and for the string $\mathcal{L}u = f$ both the load and the displacement need one. We stored a function as readings joined by segments or as coefficients on shapes, and a fixed family was exact on its members and approximate outside them. We measured error in a norm the physics chose, and two functions could be close in one norm and far apart in another without bound. We found the closest member of a family by making the error orthogonal to the family, one linear solve, and saw that the inner product was a choice. Coefficients on a basis fell at a rate set by the smoothness of the function relative to the shapes, and a jump left an overshoot that never converged away in the sup norm. Weierstrass gave density with no rate, and a unique closest member needed a closed linear subspace of a Hilbert space. L^2 had no point values and H^1 restored them in one dimension, so kernels could turn scattered readings into a curve with a norm that measures roughness and a band conditional on the model. A family of functions had a width that bounded every fixed basis at once, and particles, trained shapes, and selection dropped the fixed-basis hypothesis in three ways. Hats model every continuous function, Galerkin's method finds their weights from the equation, and nodes that move with the features need far fewer numbers than a fixed grid. The kernel prior that reconstructed the fin also solved $\mathcal{L}u = f$ for the string, which closes the loop the lecture opened with. The same representations extend to fields on a region, a membrane under a load, where the count of values grows with the dimension and H^1 no longer guarantees point values.